

Supplementary Information

CoordRep: A Canonical, Continuous, and Compositional Representation for Coordination Chemistry

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Content

Supplementary Methods	3
Supplementary Methods 1. Data sources, scope definition, and filtering rules.....	3
Supplementary Methods 2. CoordRep record grammar and sequence-interface controls.	5
Supplementary Methods 3. Molecular coordination-state construction, canonicalization, and multi-resolution identity.....	14
Supplementary Methods 4. Representation-feature comparison and attributed-graph controls	22
Supplementary Methods 5. Continuous geometry and CShM analyses	29
Supplementary Methods 6. Graph, GNN, and 3D equivariant baselines.....	33
Supplementary Methods 7. April 2025 CSD molecular and canonical periodic audits...	37
Supplementary Methods 8. Reproducibility package and file manifest	42
Supplementary Results.....	44
Supplementary Tables	46
Supplementary Figure.....	错误!未定义书签。
Source Data.....	65

Supplementary Methods

Supplementary Methods 1. Data sources, scope definition, and filtering rules

SM1.1 Evidence domains and claim boundaries

Molecular learning cohort. Molecular geometry and property analyses used a frozen 48,057-record tmQM/GFN2-xTB cohort.⁵¹ The same molecular cohort underlies the Figure 3 CShM atlases and the graph-grouped Figure 4 property benchmarks. For model comparison, records were grouped by complete atom- and bond-attributed molecular graph, and GINE,⁵⁵ SchNet,⁵⁶ ViSNet,⁵⁷ and matched CoordRep hybrids were evaluated under the same outer-fold assignments.

Full-CSD structural audit. The locally installed April 2025 CSD snapshot11 was scanned in full. Of 1,371,757 entries, 602,116 three-dimensional entries containing at least one in-domain d-block metal constitute the operational target set. All 602,116 entries yielded schema-valid audit records. Structural records were retained for 1,984,062 of 1,986,198 in-domain metal sites (99.89%) across 601,402 entries. The CSD Python API58 was used for database access and source-object reconstruction.

Molecular identity analyses. Two molecular subsets are used for analyses requiring more restrictive structural resolution. The strict mononuclear η^1 core contains 101,878 records, whereas the connectivity-supported molecular tier contains 84,453 records. The latter is used for L0–L3 family analysis; 3,491 multi-record CSD refcode families comprising 9,056 records form the main-text identity benchmark. These subsets are analysis tiers within the 602,116-entry target corpus and are not independent census denominators.

Periodic local-state analysis. Translation-resolved periodic analyses used the public 15,906-entry CSD MOF Collection.^{59,60} Canonical processing completed for 15,905 entries, 10,948 entries emitted at least one local metal-site state, and 172,332 local states were generated. These records describe periodic local first coordination spheres and their metric/stereo identities; they are not whole-framework net identifiers.

SM1.2 Data sources, versions, and redistribution policy

The tmQM-derived molecular cohort was frozen before the present revision and contains 48,057 records with GFN2-xTB geometries and the computed targets used in this work.⁵¹ All molecular model comparisons use this same frozen record set unless a smaller explicitly defined pair cohort is stated.

Crystallographic analyses used a locally installed April 2025 CSD snapshot¹¹ (version 6.00; 1,371,757 entries) accessed through the CCDC Python API,⁵⁸ version 3.6.0 used for the frozen release-wide audit. Periodic analyses used the public CSD MOF Collection^{59,60} in its supplied P1 representation.

Raw licensed CSD coordinates and structure files are not redistributed. Reproducibility materials contain only permitted derived quantities, including refcodes, scope/status labels, aggregate statistics, canonical or hashed identity keys, field-level annotations, and figure-source tables.

SM1.3 Record scopes and identity resolution

Mononuclear molecular records. Well-posed atom-resolved η^1 coordination snapshots are represented by the canonical molecular record containing metal context, ligand modules, donor relations, and continuous geometry fields. This domain supports the exact molecular canonicalization and L0–L3 identity construction described in SM3.

Expanded molecular records. Multinuclear, bridging, and haptic/ π coordination are represented through typed metal-site–donor-group incidence graphs, shared bridge objects, and connected coordination-site objects. Their source topology is audited across the full CSD target set, while exact re-expression invariance is independently tested using the multimetal and haptic template ensembles described in SM3.

Periodic local-state records. Polymeric structures are represented through translation-resolved local quotient records in which each metal–donor incidence retains the wrapped donor orbit and relative lattice translation. Periodic metric identity (MID) and stereo identity (SID) are defined separately from the molecular L0–L3 hierarchy.

CoordRep records source-resolved structural information rather than inferring chemical information absent from the source. Unresolved disorder, ambiguous collective donor assignments, missing geometry, or unresolved lattice translations

therefore reduce record resolution or generate explicit status flags rather than being silently imputed. The present implementation does not define whole-framework net identity, equivalence across alternative disorder models, dynamically averaged coordination states, or protonation/redox assignments beyond those supported by the source record.

SM1.4 Analysis denominators and subset relationships

The quantitative analyses in this work use several noninterchangeable denominators. The April 2025 CSD contains 1,371,757 total entries; 602,116 three-dimensional entries containing at least one atom in the frozen in-domain set Sc–Zn, Y–Cd, La, and Hf–Hg form the full structural-audit target set. Entries containing only Ce–Lu, actinides, or main-group metal atoms are outside this denominator. In mixed-metal entries, native contacts to out-of-domain metals are retained as typed external-center or external-donor relations. Thus, 602,116 is the predefined three-dimensional d-block target corpus, not the number of all metal-containing CSD entries. The 101,878-record strict mononuclear core and 84,453-record connectivity-supported molecular tier are nested analysis subsets. The 48,057-record tmQM cohort is an independent computed molecular dataset. The 15,906-entry CSD MOF Collection is a separate periodic collection and is not added to the full-CSD molecular denominators.

The complete census, subset relationships, and periodic counts are summarized in Supplementary Table S9 and detailed in SM7.

Supplementary Methods 2. CoordRep record grammar and sequence-interface controls

SM2.1 Scope and record-object types

CoordRep uses a shared typed vocabulary across molecular, multinuclear, haptic/ π , and periodic coordination environments, but it does not force all structural domains into one undifferentiated string or one universal chemical identifier. Three interfaces are distinguished here.

First, a compact canonical molecular string is defined for well-posed mononuclear, atom-resolved η^1 coordination snapshots. It contains the metal context, continuous shape field, donor-relation fields, and ligand dictionary. This is the sequence interface used in the masked-field and semantic-ranking experiments and is the source object for

the molecular L0–L3 hierarchy defined in SM3.

Second, release-scale crystallographic auditing uses the typed `CoordRep-Record/3` envelope. It explicitly accounts for every in-domain metal site and represents metal sites, donor groups, metal–donor incidences, direct and bridge-mediated metal relations, external-metal relations, donor-pair geometry, periodic-image information, and resolution flags. A schema-valid audit record may be resolved, partial, disorder-ambiguous, or audit-only. These are typed record outcomes, not pipeline exceptions. Multinuclear and haptic structures are represented in this envelope as incidence graphs and coordination-site objects.

Third, eligible periodic first coordination spheres can emit compact local-state records together with translation-labelled edge witnesses. Periodic MID/SID identities are derived only after the additional chemical, numerical, and crystallographic-invariance gates described in SM7. A periodic local-state record is not a whole-framework, pore, guest, net, or crystallographic-species identifier.

The sequence-interface controls in SM2.6 and SM2.7 apply to the compact molecular strings used by the corresponding sequence models. They do not imply that the release-audit JSON envelopes or periodic quotient graphs were supplied directly to those models.

SM2.2 Common fields and structural-record/identity distinction

Across record types, `CoordRep` separates metal-site attributes; local coordination counts; typed donor or coordination-site objects; metal–donor incidences; donor-pair, bridge, and metal–metal relations; geometry fields when supported by coordinates; ligand or component payloads with explicit provenance; periodic translations where applicable; and resolution, provenance, and issue fields. Electronic fields are emitted only when provided or assigned under the stated preprocessing policy. The release-wide adapter does not infer oxidation state, spin, protonation, charge, or a unique disorder realization merely to complete a record.

The release-audit envelope has the following logical structure:

```
<EntryRecord> ::=  
  <provenance> <scope_and_flags> <metal_sites> <donor_groups>  
  <coordination_incidents> <metal_relations>  
  <external_center_relations> <external_donor_relations>  
  <donor_group_relations> <donor_pair_relations>  
  <entry_resolution> <issue_codes> <record_checksum>
```

Each metal-site object has record level `local_state`, `site_object`, `topology`, or `audit_only`, together with site resolution `resolved`, `partial`, `ambiguous`, or `unresolved`. Entry-level emitted outcomes are `EMITTED_RESOLVED`, `EMITTED_PARTIAL`, `EMITTED_AMBIGUOUS`, and `EMITTED_AUDIT_ONLY`. Input or pipeline failures are written to a separate failure ledger and cannot be counted as emitted records.

A structural record and an identity key answer different questions. A structural record states which source-supported information was retained and at what resolution. A canonical identity additionally asserts equivalence under a specified transformation domain. Within the release-audit envelope, `exact_label_free` means that intrinsic record features fixed the relevant local ordering without a source-order tiebreak. If an exact intrinsic tie remains, the object is retained as `source_order_noncanonical` with an issue flag. It remains a valid, source-auditable structural record, but it is not claimed to possess an exact label-free identity under arbitrary source relabelling.

The record checksum is also not a chemical identifier. It is a SHA-256 integrity digest of the normalized structural body. Source index, refcode, release, protocol identifier, and issue bookkeeping are provenance rather than chemical-identity fields. Molecular L0–L3 identities and periodic MID/SID identities are defined separately in SM3 and SM7.

SM2.3 Mononuclear molecular grammar

The compact surface grammar is used for well-posed mononuclear η^1 records. Its

simplified form is:

```
<molecular_record> ::= <metal_block> <shape_block_or_status>
                        <relation_block>* <ligand_dictionary>

<metal_block> ::= "[Metal:" <element>
                  ("|ox:" <oxidation_state>)?
                  ("|d:d" <nonnegative_integer>)?
                  "|CN:" <positive_integer> "]"

<shape_block_or_status> ::=
    "<ShapeBest:" <shape_label> "|Class:" <shape_class>
    "|Delta:" <delta_bin> "|V:" <number_list> ">"
    | "<ShapeStatus:unsupported|Reason:" <reason>
    "|CN:" <positive_integer> ">"

<relation_block> ::= "{trans:" <donor_ref> "--" <donor_ref> "}"
                    | "{cis:" <donor_ref> "--" <donor_ref> "}"
                    | "{fm:" <fac_or_mer> "}"

<donor_ref> ::= "L" <ligand_index> ":" <element> ":" <donor_rank>

<ligand_entry> ::= "L" <ligand_index> "="
                  ("SMILES:" <canonical_smiles>
                   | "FORMULA:" <formula_payload>)
```

The fixed surface order is metal $\rightarrow$ shape $\rightarrow$ donor relations $\rightarrow$ ligand dictionary. Optional oxidation-state and d-count fields are absent rather than imputed when unavailable. CN is the number of accepted atom-resolved donor sites in the molecular first sphere. Display line breaks are not part of the grammar.

The shape block stores the best reference label, a reporting class, a discretized top-two competition field, and the CShM vector in the frozen reference order for that coordination number. CShM calculation and bin definitions are specified in SM5; this section defines only their serialization. A donor pointer such as L2:N:1 is local to the canonical ligand dictionary and is not a source atom label. Relation endpoints are serialized only after donor and ligand canonicalization.

SMILES: denotes a connectivity-resolved ligand payload under the stated graph policy. FORMULA: is an explicit composition-only fallback and cannot support the same

constitutional claim. The molecular identity consequences of this provenance distinction are defined in SM3.8.

SM2.4 Multinuclear record graphs

Finite multicenter structures are represented as typed metal-site–donor-group incidence graphs. Each in-domain d-block center defines a metal-site node, and each atom-resolved donor or collective coordination site defines a donor-group node. A `CoordinationIncidence` records one native metal–donor-member contact, including the metal, donor group, donor ordinal, donor element, σ/π role, source bond type, distance when available, and periodic image delta when applicable.

A donor group shared by two or more in-domain metal centers is stored once, with its complete target-metal set and `bridge_degree`. It is not duplicated as unrelated terminal ligands for separate local spheres. One incidence is emitted for every native metal–donor-member contact, and a `shared_donor` metal relation points back to the shared group. When two distinct coordination-site objects overlap on source atoms, the overlap is retained as a donor-group relation rather than collapsed; this preserves η/σ and η/η membership overlap.

Direct metal–metal edges are emitted only when supported by a native source bond and remain distinct from bridge-mediated connectivity. A bonded metal outside the predefined audit domain is retained as an external-center relation rather than promoted to an independently audited in-domain site. A donor atom shared with an out-of-domain metal is represented by an external-donor relation. These typed objects preserve heterometallic context without changing the frozen audit denominator.

For each metal site, `cn_site` counts incident coordination-site objects and `cn_atom` counts physical metal–donor incidences. The record also retains direct-metal degree, external-metal degree, external-bridge count, donor composition, and donor-pair geometry. A crystal entry may contain disconnected metal-centered components; connectivity is never forced. If intrinsic ordering signatures leave an exact tie, the record remains auditable but is marked `source_order_noncanonical`. Exact finite-graph canonicalization and its validated domain are described in SM3.5 and SM3.10.

SM2.5 Haptic/ π and periodic local-state records

Haptic or π coordination is represented as a coordination-site object rather than as an unstructured set of independent η^1 donors. For each target metal, donor atoms carrying native π or Delocalised metal–donor roles are collected and connected components are determined in the source nonmetal graph. A connected component containing 2–12 donor atoms is emitted as a `pi_fragment` with explicit membership, donor elements, topology witnesses, target metals, bond roles, and hapticity. A singleton remains an atom-level site. A component with more than 12 members is retained conservatively as a `candidate_atom_set`; definite η is withheld and an ambiguity flag is emitted.

For a resolved collective site, centroid-derived local-geometry summaries are calculated from the source-supported member atoms incident on the target metal. `cn_site` counts the collective site once, `cn_atom` counts its physical metal–donor contacts, and `eta_sum` sums resolved hapticities. Thus, $\text{Fe}(\eta^5\text{-Cp})_2$ has `CNsite` = 2 and `eta_sum` = 10; it is not represented as an ordinary η^1 , `CN` = 10 sphere.

For a periodic record, every physical metal–donor incidence additionally carries an integer image-difference vector $\tau \in \mathbb{Z}^3$ between the source lifts of the metal and donor. The donor group stores a wrapped periodic-orbit witness separately. Contacts to the same wrapped donor orbit are not merged when their τ values differ. If the source object does not support an image assignment, `image_delta_resolved` is false and the numerical placeholder is not interpreted as a resolved zero-translation edge.

The translation-labelled record retains the periodic information supplied by the source graph; it is not by itself an identity invariant to arbitrary changes of unit-cell setting or origin. Canonical quotient construction, crystallographic re-expression tests, and MID/SID identities are described in SM7.

SM2.6 Sequence-interface tokenization and metal/CN factorization

For the molecular sequence experiments, each compact string was divided into intact metal, shape, relation, and ligand blocks. Structured headers and complete relation expressions were preserved as recognizable token classes, while ligand payloads and the remaining surface context used the same character-level policy in both

controlled variants.

The composite interface represents the entire leading header as one structured token, whereas the factorized interface exposes its fields independently:

`composite: [Metal:Pt|ox:+2|d:d8|CN:4]`

`factorized: [Metal:Pt] [ox:+2] [d:d8] [CN:4]`

This changes only the sequence interface. Model output is reassembled into the canonical compact header before validation or identity construction. The underlying record grammar, shape fields, and CoordRep-ID definitions are unchanged.

The leakage-controlled comparison began with 51,250 parsed records. Grouping by sorted ligand strings and denticities gave 41,296 training, 4,891 validation, and 5,063 test records. Exact CoordRep and ligand-signature overlap were both zero across splits, and vocabularies were fitted on the training split only. Both interfaces used the same 558,944-parameter, three-layer masked Transformer (hidden size 96, six heads, feed-forward size 384, maximum length 256, vocabulary capacity 2,048), AdamW with learning rate 4×10^{-4} and weight decay 0.01, cosine decay, batch size 192, 12 epochs, BF16 arithmetic, and seeds 11, 22, and 33. The lowest-validation-loss checkpoint was used.

In the equal-information intervention, the complete leading header was hidden in both variants. Factorization increased metal Top-1 recovery from $60.46 \pm 0.15\%$ to $66.85 \pm 0.18\%$ and joint metal-plus-CN recovery from $54.98 \pm 0.15\%$ to $60.27 \pm 0.17\%$. The paired improvements were +6.39 percentage points (95% bootstrap CI, +5.48 to +7.28) and +5.29 points (+4.38 to +6.19), respectively. Exact full-header recovery increased by +3.38 points (+2.48 to +4.30), whereas CN recovery changed by +0.22 points (−0.16 to +0.63). Factorization also permits metal-only and CN-only masking, which is not expressible for one fused token. These results demonstrate field addressability, not a change to chemical identity (Supplementary Table S7).

SM2.7 Field-block order stress test

A paired inference-time stress test evaluated whether frozen sequence models depended on canonical block order; no checkpoint was retrained. The canonical

sequence and fixed global permutation were:

canonical: metal → shape → relations → ligands

fixed: ligands → relations → shape → metal

They were compared with record-specific deterministic permutations derived from SHA-256(record identifier | order seed) for seeds 11, 22, and 33. Canonical end truncation to 512 tokens was applied before permutation. Every comparison therefore retained the same post-truncation token multiset, masked target, test identifier, tokenizer, checkpoint, and numerical settings. Delimiters moved with their field blocks. Each authentic semantic record and all of its decoys received the same permutation.

The masked-field evaluation contained 4,000 donor targets from 2,000 complexes and 1,966 structure-field targets. Semantic ranking contained 1,450 strict groups and 10,877 authentic-decoy pairs; its stereo-hard subset contained 960 groups and 2,254 pairs. Uncertainty used 10,000 cluster-bootstrap resamples at the complex or semantic-group level.

Donor Top-1 recovery was 85.575%, 84.300%, and 84.958% for canonical, fixed, and record-specific order, respectively; structure-field recovery was 97.406%, 94.151%, and 96.253%. Strict AUROC was 0.7981, 0.5589, and 0.6706, and stereo-hard AUROC was 0.9475, 0.5917, and 0.7910 (Supplementary Table S8). Canonical field order is therefore part of the learned sequence interface. This stress test measures positional sensitivity of checkpoints trained on canonical strings; it does not establish tokenizer permutation invariance or redefine the parser, record, or identity hierarchy.

SM2.8 Record validation and round-trip requirements

For a molecular record used as an exact CoordRep state, validation requires successful parsing of all delimiters; field-type and enumeration validity; valid donor and ligand references; consistency between CN and the accepted donor set; absence of dangling labels; canonicalize–serialize round-trip equality; and regeneration of the same L0–L3 keys from the round-tripped structural record. Exact canonical invariance is an additional test in SM3 and is not inferred from syntax alone.

For CoordRep–Record/3, schema validation requires that the declared metal-site count equals the emitted site list; local labels are unique; every incidence and relation references existing objects; donor ordinals, elements, bond roles, and three-component

integer image deltas are internally consistent; donor-group kind, hapticity, target set, bridge degree, and external-contact counts agree with the incidences; and `cn_site`, `cn_atom`, `eta_sum`, composition, degrees, and donor-pair counts recompute for every metal site. Relation endpoint order is normalized, duplicates are rejected, unresolved geometry cannot carry a numerical angle, and the record checksum must recompute from the normalized structural body.

A controlled schema failure is written to the failure ledger and is not relabelled as a partial chemical record. Conversely, missing coordinates, disorder, an unresolved translation, or an ambiguous collective π component can legitimately lower the information resolution while leaving the typed envelope schema-valid. Schema validation tests internal consistency; independent rereading of the native crystallographic object tests source fidelity. Release-wide source-fidelity methods and results are reported in SM7.

SM2.9 Machine-readable serialization and display policy

Compact molecular interfaces are stored as unwrapped strings. Release-audit envelopes are stored as normalized JSON with recursively ordered mapping keys, typed lists, ASCII-safe escaping, no insignificant whitespace in the canonical byte stream, and no nonfinite numerical values. Schema version and protocol identifier accompany each object so that records produced under different conventions are not silently mixed.

Line breaks, indentation, colors, and chemical abbreviations inserted in Table 1, figures, or the Supporting Information are presentation features without semantic meaning. Parsing, round-trip validation, checksums, and identity construction use the complete machine representation. Block-permuted strings from SM2.7 are experimental sequence inputs and are not stored as canonical records.

Recognizable aliases such as `en`, `Cp`, or `BTC` may be used for display only when mapped to the stored payload. `SMILES`: marks connectivity-resolved provenance under the stated graph policy, whereas `FORMULA`: marks composition only. Missing information is omitted, stored as null, or accompanied by a typed status; it is never silently imputed to make a record appear complete. Short hashes may be shown for layout, but equality is determined from the complete canonical string or complete key defined in SM3.

Supplementary Methods 3. Molecular coordination-state construction, canonicalization, and multi-resolution identity

SM3.1 Scope and identity levels

This section defines molecular coordination-state construction, exact canonicalization, and the four molecular identity levels L0–L3. Record grammar and sequence controls are in SM2; representation-family and attributed Weisfeiler–Lehman comparisons are in SM4; coordinate-space CShM calculation is in SM5; and the release-wide CSD audit and periodic MID/SID identities are in SM7.

Canonicalization and identity resolution are different operations. Canonicalization removes nuisance variation among descriptions of one encoded record, including rigid translation, proper rotation, input container order, ligand traversal, and source identifiers within the validated domain. Identity resolution determines which canonical fields must agree. L0 retains the complete serialized coordination-state record at the adopted numerical precision; L1–L3 progressively remove geometry and stereochemical detail.

An L0 match therefore means equality of two serialized coordination-state records. It is not a claim that their raw Cartesian coordinates, complete chemical species, protonation states, disorder models, or all source annotations are identical. Molecular L0–L3 keys are distinct from the periodic MID/SID comparison axis.

SM3.2 Molecular coordination-state record object

A molecular coordination-state record is constructed from a snapshot containing a designated metal center, its assigned first-sphere donors, a graph from which donor-bearing ligand components can be extracted, a donor-to-ligand mapping, donor coordinates for geometry and relation calculations, and metal-state fields when supported by the input policy.

The validated mononuclear core requires one in-domain metal center and an atom-resolved η^1 interpretation. CN is the number of retained first-sphere donor atoms. The record is well posed only when the metal, donor set, ligand partition, and donor-to-ligand mapping are uniquely defined under the adopted rules. Missing three-dimensional information, unresolved first-sphere membership, severe disorder, or

unsupported ligand construction is not repaired by inventing an assignment; the information is rejected from the exact domain or retained at an explicitly lower audit resolution.

The object represents one experimentally or computationally resolved coordination state. It is not a whole-species identifier and does not establish equivalence among alternative protonation, oxidation, tautomeric, disorder, or bonding models.

SM3.3 Internal typed representation

Before serialization, an accepted molecular snapshot is converted to a typed `CoordComplex` object:

`CoordComplex`

metal: element, oxidation, d_count

graph: metal_index, donor_indices, donor_to_ligand_map,

metal-donor bond fields when available

ligands[]: payload, provenance, connectivity_status,

attachment_set_key, donor_attachment_keys,

donor_rank_orbits, donor_elements, denticity,

hapticity, charge when available

shape: CN, ordered reference labels, coordinate-space CShM vector

constraints: trans pairs, cis pairs, supported fac/mer annotation

source_id; issues[]

Ligand provenance is explicit. A formula fallback is distinguishable from a connectivity-resolved ligand and cannot silently acquire graph semantics. Unsupported shape calculation produces a typed shape-status block rather than an invented numerical value. Source or global atom indices are retained only as provenance and are not used in any canonical ligand-, donor-, or identity-ordering key.

Canonicalization is performed on a copy of the typed object. Once ligand and donor references are canonical, the serializer emits metal, shape, relation, and ligand blocks in the SM2 grammar. The molecular L0–L3 keys are derived only from this canonical unwrapped record.

SM3.4 Ligand and donor-site construction

After first-sphere assignment, the designated metal is removed from the molecular graph and connected components are calculated on the remainder. Each component containing at least one retained donor becomes a ligand module. A component without

a retained donor is excluded from the coordination-state ligand dictionary unless retained separately as provenance.

The implementation attempts to generate a canonical ligand SMILES under a fixed graph and sanitization policy. If a usable graph is unavailable or sanitization fails, the payload is tagged `FORMULA` with unsupported-connectivity status. Nominal SMILES based on inferred rather than supplied bond orders retains its corresponding confidence status. These distinctions propagate to the identity keys.

For a graph-supported ligand, the implementation constructs a source-index-free attachment-set key by canonically coloring all coordinated atoms in the isolated ligand graph. A donor-specific attachment key applies an additional color to the queried donor while retaining the complete attachment-set coloring. Equal donor-specific keys define donor orbits under the stabilizer of the complete coordinated-atom set. Attachment identity is retained in the serialized ligand payload and the internal ordering key. A repeated ligand payload for which attachment identity is unavailable fails closed rather than being treated as freely exchangeable.

Every atom-resolved donor is addressed as $L_k:E:r$, where L_k is the canonical ligand identifier, E is the donor element, and r is its local donor rank under the fixed adapter convention. Haptic/ π coordination is not expanded into unrelated η^1 donors; the expanded grammar uses the coordination-site object defined in SM2.5.

SM3.5 Chemical, attachment-aware donor signatures and exact tie resolution

Canonical ordering first uses chemical and ligand-local attachment information and then applies rotation-invariant metric refinement. Each graph-supported ligand module L receives a joint attachment-set key A_i , defined by a canonical colored ligand graph in which all coordinated atoms are marked. For donor i , a donor-specific attachment key a_i additionally marks the queried donor while retaining the full coordinated-atom set.

$$f_i = [E_i, a_i, \hat{d}M_i, \text{sort}(\{\hat{d}_{ij} : j \neq i\})].$$

Here E_i is the donor element, a_i is the source-index-free donor-attachment key, $\hat{d}M_i$ is the quantized metal–donor distance, and the final term is the sorted quantized donor–donor distance row. These fields are invariant to proper rigid motion and contain no source or global atom index.

Ligand instances are treated as exchangeable only when their payload fields, joint attachment-set keys, and multisets of donor-specific attachment keys agree. Donor ranks can remain tied only within the same element, attachment orbit, and quantized geometric-refinement class. These rules prevent repeated ligand payloads from being exchanged across nonautomorphic attachment patterns.

When residual classes remain, the implementation enumerates the Cartesian product of all legal ligand-instance and donor-rank permutations, serializes the complete record for every candidate, and selects the lexicographically minimum whole record. Candidate counts are checked before enumeration; unsupported or metadata-deficient cases are rejected explicitly rather than truncated, approximated, or resolved by source order.

Finite multicenter record graphs use exact individualization–refinement for at most 12 metal nodes. Refinement uses typed metal edges and the complete metal–site–ligand incidence context; discrete leaves are scored by a source-label-free complete code, and the lexicographically smallest code defines the metal order.

SM3.6 Stereochemical relation construction

Relations are constructed after canonical donor pointers have been assigned. For donor coordinates r_i and r_j and metal coordinate r_M ,

$$\theta_{i,j} = \arccos[\frac{(r_i - r_M) \cdot (r_j - r_M)}{(|r_i - r_M| |r_j - r_M|)}].$$

The molecular-core policy assigns *trans* when $\theta_{i,j} \geq 165^\circ$ and *cis* when $75^\circ \leq \theta_{i,j} \leq 105^\circ$; other pairs remain outside those classes. Supported fac/mer patterns are emitted as {fm:fac} or {fm:mer}. A record that does not satisfy a supported pattern receives no forced fac/mer label.

Within each relation, donor pointers are ordered lexicographically. Sorted *trans* tokens precede sorted *cis* tokens and the fac/mer field. The complete pointer-resolved map is retained in L0. L1 retains only the numbers of *trans* and *cis* relations and the fac/mer label; it does not retain the complete donor-pair mapping. L2 and L3 omit relation fields.

SM3.7 Continuous shape-field serialization

CShM values are calculated by the corrected coordinate-space procedure in SM5. The metal is treated as the distinguished center, admissible donor-to-reference assignments are enumerated, and proper rotation and uniform scale are optimized. Legacy greedy donor-distance-matrix scores are not used as CShM values.

The surface block is:

```
<ShapeBest:LABEL|Class:CLASS|Delta:d|V:v1,v2,...>
```

v follows a fixed reference order: L for CN = 2; TP for CN = 3; Td, SP for CN = 4; TBP, SPY for CN = 5; and Oh, TPr for CN = 6. Values are calculated before rounding and serialized to two decimal places. Best-CShM class is *ideal* for $S_best < 1$, *good* for $1 \leq S_best < 3$, *dist* for $3 \leq S_best < 8$, and *irreg* otherwise. With $\Delta S = S_second - S_best$, Δ is 0 for $\Delta S < 0.5$, 1 for $0.5 \leq \Delta S < 2$, 2 for $2 \leq \Delta S < 5$, and 3 otherwise. L1 additionally marks a shape boundary when $\Delta S < 1.0$. Threshold sensitivity is reported in Supplementary Figure S1.

If no implemented reference set or usable coordinate vector exists, the serializer emits `<ShapeStatus:unsupported|Reason:...|CN:n>` rather than an imputed geometry. CShM values describe static geometric proximity to references; they do not by themselves establish a reaction coordinate or dynamic interconversion.

SM3.8 L0–L3 CoordRep-ID hierarchy

The molecular hierarchy is derived deterministically from the canonical record (Supplementary Table S11).

L0 – CoordRep-State. L0 is the complete canonical unwrapped CoordRep string itself, not a hash. It retains metal, available oxidation and d count, CN, the rounded CShM vector and shape fields, complete donor-pointer relations, and provenance-tagged ligand payloads. L0 is exact for the serialized fields and precision; it is not a lossless Cartesian-coordinate identifier.

L1 – ShapeID.

metal | oxidation | CN | boundary-aware binned-shape token |
relation-class/count signature | sorted ligand identities

L1 omits exact CShM decimals and donor-pair pointers but retains best/second shape, best-CShM and ΔS bins, boundary state, oxidation, CN, counts/types of relations, and ligand identities. The d count is not an L1 field.

L2 – TopoID.

metal | CN | best-shape label | sorted ligand identities

L2 omits oxidation, shape-bin and boundary detail, and stereochemical relations. The molecular implementation contains no additional donor-map or metal-graph signature at L2; its safe interpretation is a best-shape coordination family with ligand identities.

L3 – ConnID.

```
metal | CN | sorted provenance-tagged ligand payloads |  
connectivity-support status
```

L3 removes geometry. A formal connectivity-supported ConnID requires SMILES provenance for every ligand. Formula-containing and legacy-untagged records retain diagnostic L3 strings with explicit unsupported status and cannot be interpreted as validated ligand-connectivity identities. L3 is not a whole-species InChI or a complete molecular graph identifier.

For validated, connectivity-supported records for which all four keys are formally defined, equality is nested: an L0 match implies L1, L2, and L3 matches; an L1 match implies L2 and L3 matches; and an L2 match implies an L3 match. Explicit CN in L1 is required to preserve this nesting when shape status is unsupported.

SM3.9 Identity-key and hash policy

The complete text strings defined above are authoritative. L0_StateKey is the full canonical CoordRep string; L1_ShapeID, L2_TopoID, and L3_ConnID are the complete derived key strings. SHA-256 values are storage and indexing aids, not independent chemical definitions.

For family analysis, complete keys are UTF-8 encoded and hashed with full SHA-256. Display line wrapping, typography, and abbreviation are absent from the hashed input. A short digest shown in a schematic is only a prefix and is never used to decide equality or group records. The audit-record checksum in SM2.2 and the molecular identity hashes serve different purposes and are not interchangeable.

Serializer, parser, CShM reference, binning, and identity-key versions are frozen in the provenance. Keys from different versions are not assumed equivalent without an explicit migration test. In any collision or audit dispute, the complete canonical string is authoritative.

SM3.10 Canonical-invariance and exact-orbit stress tests

The general end-to-end challenge used a fixed population of 6,427 eligible unique molecular graphs after requiring one transition-metal center, CN = 4–6, successful tmQMg/PBE record construction, no validation issue, graph-supported ligand payloads, and verified coordinate-source hashes. A graph-

disjoint cohort of 1,000 snapshots was selected without reference to canonicalization outcomes (CN = 4/5/6: 334/333/333; 200 records from each frozen outer fold). For every snapshot, 100 deterministic trials were generated in each of three arms: proper rigid motion, consistent atom/connection-table reindexing, and their combination. CoordRep reproduced 100,000/100,000 variants and collapsed 1,000/1,000 structures in each arm, corresponding to 300,000/300,000 exact variant matches overall. Control outcomes are reported in Supplementary Table S2A.

A separate legal-orbit challenge tested whether invariant-key sorting is sufficient when chemically admissible residual ties remain. Of the 6,427 eligible unique graphs, 5,173 contained at least one non-singleton legal ligand-instance or donor-rank orbit. A fixed graph-disjoint subset of 1,000 structures (CN = 4/5/6: 716/74/210) underwent 100 atom reindexings per structure. Signature-only sorting reproduced 64,834/100,000 variants and completely collapsed 380/1,000 structures. Full legal-orbit enumeration followed by minimum-whole-record selection reproduced 100,000/100,000 variants and collapsed 1,000/1,000 structures. All exact strings passed independent grammar round-trip validation; candidate counts were stable, and no variant or fatal error occurred. Candidate-count and timing distributions are reported in Supplementary Figure S2.

A nonautomorphic adjacent-versus-opposite attachment-set control remained distinct under exhaustive legal relabeling, with zero false merges. A repeated-payload record lacking the attachment metadata required to establish exchangeability was rejected explicitly. Idealized fac- and mer-[Co(NH₃)₃Cl₃] models provided an independent specificity control: all displayed non-relation fields were fixed, whereas the trans-relation inventories were $3 \times \text{N-Cl}$ and $\text{N-N} + \text{N-Cl} + \text{Cl-Cl}$, respectively, yielding distinct CoordRep-States without using fac/mer labels as encoded inputs. Earlier exact-tie regressions also remained satisfied: 4/4 CN = 4 orders, 8/8 CN = 6 orders, 24/24 Cu₄-cycle metal orders, 5,500/5,500 multimetal variants, and 3,300/3,300 haptic variants converged as specified, while a nonisomorphic Cu₄ path remained distinct (Supplementary Table S2B). The expanded templates test information present in the typed record abstraction and are not CSD prevalence samples.

These challenges test fixed-snapshot serialization. Actual coordinate changes can legitimately alter rounded CShM or donor-relation fields and therefore change L0.

SM3.11 Molecular family benchmark and XEYVEC case

The corrected connectivity-supported molecular tier contains 84,453 records. Refcode families with at least two accepted records yielded 3,491 families and 9,056

records. For family f , identity level ℓ , and key multiplicity $n_{fk\ell}$, pairwise agreement was

$$A_{f\ell} = [\sum_k C(n_{fk\ell}, 2)] / C(n_f, 2).$$

The reported value is the unweighted mean of $A_{f\ell}$ over families; complete concordance means $A_{f\ell} = 1$. Mean pairwise agreement increased from 11.2% at L0 to 83.5%, 96.6%, and 97.4% at L1–L3. Complete concordance occurred in 356, 2,837, 3,351, and 3,379 families, respectively.

The XEYVEC Cu(II), CN = 4 family is the largest corrected same-L3 family occupying more than one best-shape basin. Five $[\text{CuCl}_4]^{2-}$ determinations form two reproducible intermediate regions of the SP–Td plane. XEYVEC, XEYVEC02, and XEYVEC04 are yellow C2/c polymorphs and are Td-leaning: $[S(\text{Td}), S(\text{SP})] = [6.92, 11.76]$, $[7.42, 11.14]$, and $[7.50, 11.05]$. XEYVEC01 and XEYVEC03 are green P–1 polymorphs and are SP-leaning: $[11.84, 6.82]$ and $[11.85, 6.82]$.

The family yields five L0 states, two L1 ShapeIDs, two L2 TopoIDs, and one connectivity-supported L3 family. Because all best CShM values remain finite and intermediate, neither group is an ideal SP or Td endpoint. XEYVEC demonstrates static polymorph-associated geometry basins within one coordination constitution; it does not establish a dynamic interconversion pathway.

SM3.12 Scope, source fidelity, and unresolved states

The strongest canonical claim here is exact serialization of fields present in a well-posed geometry-resolved record within the validated equivalence domain. Canonicalization cannot recover distinctions absent from the source or adapter. It does not independently determine protonation, oxidation, spin, disorder realization, bond order, ligand constitution, or stereochemistry that was not encoded.

Formal L3 connectivity identity requires SMILES-resolved provenance. Formula fallbacks and legacy payloads remain searchable diagnostic records but are excluded from the 84,453-record connectivity-supported family benchmark. The exact mononuclear search is validated for $\text{CN} \leq 6$, and oversized cases are rejected. Internally tied donors remain outside arbitrary relabelling equivalence unless canonical ligand-graph atom orbits are supplied.

The release-wide adapter distinguishes label-free exact ordering from source-order-tied audit output. A structurally resolved record can still contain an unresolved ordering tie, and a label-free record can still inherit source disorder or missing annotations. Missing geometry, unavailable CShM references, ambiguous collective π sites, disorder alternatives, and unresolved periodic translations are retained as typed issues or lower-resolution outcomes rather than silently imputed.

Across the April 2025 full-CSD audit, all 602,116 target entries emitted schema-valid audit envelopes and all 602,116 passed the independent native-object source-fidelity comparison. Every in-domain metal site was structurally recorded in 601,402 entries (99.8814%); lower-resolution and audit-only outcomes remained explicit. These are source-record and structural-coverage results, not a claim that every entry received an exact molecular CoordRep-ID. Their construction, complete ledger, and periodic identity analyses are reported in SM7 and Supplementary Tables S9, S12, and S13.

Accordingly, CoordRep-State is an auditable identity of an encoded coordination record, not an InChI-like identifier for a complete species. Whole-framework net identity, equivalence among alternative disorder or bonding models, dynamically averaged states, and cross-version identity without an explicit migration remain outside the present scope.

Supplementary Methods 4. Representation-feature comparison and attributed-graph controls

SM4.1 Purpose of the Coordination Identity Challenge

The representation-feature comparison evaluates which structural operations are explicitly defined by each representation family. It is not a predictive-model benchmark. A downstream model may learn additional distinctions from coordinates or graphs, whereas the present comparison asks whether the representation itself exposes a stable, directly accessible object for identity, geometry, stereochemistry, validation, or family linking.

This distinction is important because many representations can be useful for a subset of coordination-chemistry tasks while leaving other operations undefined. For

example, a raw coordinate tensor contains geometric information but does not by itself define a canonical record identity. A graph hash can remove atom-order dependence at the connectivity level but does not define continuous coordination shape competition. A CShM vector encodes continuous geometry but does not encode ligand identity or stereochemical relation fields. The Coordination Identity Challenge therefore evaluates operation coverage, not whether a downstream model could potentially learn the same distinction from additional data.

SM4.2 Operations evaluated

These are the operations reported in Supplementary Table S1.

Canonical record identity. A representation supports canonical record identity if it maps equivalent inputs for the same coordination snapshot to a reproducible record or key, independent of nuisance variation such as atom ordering, file traversal order, or coordinate-frame orientation.

Coordination stereochemistry. A representation supports coordination stereochemistry if it explicitly distinguishes donor-site relations such as cis/trans and fac/mer arrangements, rather than relying only on ligand composition or graph connectivity.

Continuous or boundary-aware geometry. A representation supports continuous or boundary-aware geometry if it can encode graded distortion and near-competing shape assignments, rather than forcing each coordination environment into a single discrete geometry label.

Family linking across geometry states. A representation supports family linking if it can connect related records that share a connectivity-level coordination identity while preserving differences in geometry-resolved states.

Validation-ready record. A representation is validation-ready if it defines a parseable record with explicit grammar or structural constraints such that invalid records can be detected systematically.

Multi-resolution identity. A representation supports multi-resolution identity if it provides more than one identity layer, allowing exact snapshot matching and coarser family-level retrieval to coexist.

SM4.3 Scoring categories

Each representation family was assigned one of three scores for each operation.

Full. The operation is explicitly defined by the representation within its stated scope. A user can perform the operation directly from the representation or its specified identity keys without adding a separate task-specific model or manual rule set.

Partial. The representation contains some information relevant to the operation, but the operation is incomplete, restricted to a subset of cases, or requires additional interpretation. Partial support includes cases where the information is implicitly present but not exposed as a stable record field.

Not supported. The representation does not define the operation in a usable way, or the operation requires external information not present in the representation. The scoring is intentionally conservative. A representation is not marked “Full” merely because a downstream model or additional program could infer the missing operation from other data. The question is whether the representation itself defines the operation.

SM4.4 Representation families evaluated

The comparison was performed at the level of representation families, not individual software packages. This avoids overinterpreting implementation-specific behavior and focuses on the operations that each representation type is designed to support. The evaluated families were:

- ♦ conventional molecular strings;
- ♦ TMC structure-to-SMILES workflows;
- ♦ TMC descriptor or featurization schemes;
- ♦ stereoisomer or conformer workflows;
- ♦ m-SMILES / MetalloGen-style notations;
- ♦ T-REX trans-pair notation;
- ♦ CShM-only or coordinate-based geometry descriptors;
- ♦ atom-attributed graph hashes / Weisfeiler–Lehman²⁴-type graph keys;
- ♦ CoordRep molecular and periodic record framework.

SM4.5 Rationale for each representation family

Conventional molecular strings. Conventional molecular strings such as SMILES and InChI provide compact identifiers for many covalent molecules, and they may support partial canonicalization and limited stereochemical distinction. However, they were not designed to encode continuous metal-centered geometry, donor-site relation maps, or coordination-specific identity layers. They are therefore partial for record identity and stereochemistry, but do not provide continuous boundary geometry or multi-resolution coordination identity.

TMC structure-to-SMILES workflows. Structure-to-SMILES workflows for transition-metal complexes improve the conversion of three-dimensional TMC structures into model-ready strings. These workflows can support practical parsing and graph conversion, but their primary goal is not to define a multi-resolution coordination-record identity. They therefore partially support canonical record construction and stereochemistry for supported cases, but continuous geometry and family-level identity remain outside their core definition.

TMC descriptor or featurization schemes. Descriptor and featurization schemes are useful for predictive modeling and structure–property analysis. They often encode metal identity, ligand descriptors, or geometry-related numerical features. However, they typically do not define a parseable canonical record, a stereochemical relation map, or identity keys for database-scale linking. They are therefore partial for some geometry or stereochemistry-related information but do not support record identity or multi-resolution identity.

Stereoisomer or conformer workflows. Stereoisomer and conformer workflows are strong for enumerating or generating chemically plausible configurations in supported classes. They may fully support certain stereochemical distinctions within their target domain. However, they are not generally designed as database-scale canonical record languages with continuous boundary geometry, validator-compatible grammar, and L0–L3 identity abstraction. Their scoring is therefore full for supported stereochemical operations but partial or unsupported for broader record-level operations.

m-SMILES / MetalloGen-style notations. m-SMILES and related notations

extend molecular strings toward coordination-aware conformer generation. They provide a useful ligand/metal description and can be suitable for generation workflows. However, they do not by themselves define continuous CShM geometry fields, explicit boundary-state reporting, or a multi-resolution identity hierarchy. They therefore receive partial scores for canonicalization, stereochemistry, geometry, family linking, and validation-ready representation, but not full support for multi-resolution identity.

T-REX trans-pair notation. T-REX-style trans-pair notation directly addresses geometry-resolved transition-metal-complex representation and explicitly encodes trans-pair relationships. It therefore provides strong support for canonical identity and coordination stereochemistry within its supported domain. However, it does not primarily define continuous CShM shape competition or an L0–L3 identity hierarchy spanning exact state identity to connectivity-level family linking. It is therefore full for canonical identity and stereochemistry, but partial or unsupported for the remaining operations.

CShM-only or coordinate-based geometry descriptors. CShM and coordinate-based descriptors provide strong continuous geometry information. They are therefore full for continuous or boundary-aware geometry when the relevant shape set is defined. However, by themselves they do not encode ligand identity, donor-site relation fields, grammar validation, or family-level connectivity identity. They are therefore not record languages.

Atom-attributed graph hashes / WL-type graph keys. Deterministic attributed graph hashes provide atom-order-invariant identity for a specified molecular graph and therefore serve as strong connectivity controls. Stereochemical information can also be distinguished when the relevant relations are explicitly supplied as graph attributes. However, the attributed graph alone does not encode continuous CShM coordinates, near-boundary shape competition, or the geometry-resolved L0–L3 identity hierarchy used here.

CoordRep. CoordRep defines a canonical, field-addressable record for coordination states. Its molecular records expose metal context, continuous CShM geometry, donor relations, ligand/site identity, and validation status, while the L0–L3 hierarchy separates the finest encoded coordination-state equality at the stated serialization precision from progressively broader family identity. The expanded record

grammar additionally represents shared bridges, multinuclear site topology, haptic/ π coordination, and lattice-translation-resolved periodic contacts. Periodic local states use complementary MID/SID keys as a separate comparison axis rather than as additional molecular L0–L3 levels. Within these stated scopes, CoordRep directly supports canonical record identity, explicit coordination stereochemistry, continuous geometry, validation-ready records, and resolution-aware identity comparison.

SM4.6 Attributed Weisfeiler–Lehman control construction

➤ The attributed Weisfeiler–Lehman control was constructed on the frozen current-CSD intersection. The node labels comprised atomic number, formal charge, CCDC Sybyl type, atom chirality, CCDC metal flag, target-metal flag, and direct-donor flag. Edge labels comprised CCDC bond type/order, Sybyl bond type, E/Z stereochemical state, and direct metal–donor status. A NetworkX 1-WL refinement was applied for six iterations, and the final multiset digest was stored as a 32-byte key. Radius-3 and radius-10 sensitivity checks were included in the audit. The common intersection contained 2,112 records from 752 families after current-CSD reacquisition. The relation-augmented control inserted explicit donor-pair relation nodes for cis angles of 75–105° and trans angles of at least 165°. The construction follows the Weisfeiler–Lehman graph-kernel framework (ref 24).

The member and refcode-family manifest was frozen before construction of the current graph control and contained 2,172 records in 765 multi-entry families. Molecular structures were then reacquired by refcode from the April 2025 CSD installation. Reacquisition succeeded for 2,123 records, and donor-count agreement with the frozen records was 2,123/2,123; 49 refcodes were unavailable. Requiring complete keys for every compared method produced the common 2,112-record, 752-family intersection reported in Supplementary Table S10. This analysis is a current-release reacquisition and coverage audit, not a prospective temporal-generalization or release-shift benchmark.

SM4.7 Challenge-pair construction for audit examples

In addition to the representation-family scoring above, we constructed representative challenge pairs to audit whether CoordRep operations behave as

intended. These challenge pairs are not used as predictive-model benchmarks; they are diagnostic examples for record-level operations.

The challenge-pair categories are:

Invariance pairs. Pairs that represent the same coordination snapshot under changed atom ordering, coordinate-frame transformation, or equivalent file traversal. Expected behavior: L0 remains identical.

Stereochemical separation pairs. Pairs that retain the same atom- and bond-attributed molecular graph where required by the test but differ in explicit donor-site relations or resolved coordination state. Expected behavior: graph-level identity may remain unchanged while the relevant CoordRep state or relation field differs.

Family-linking pairs. Pairs that share coarse connectivity-level identity but differ in geometry-resolved state. Expected behavior: L0 differs, but L3 matches.

Boundary-geometry examples. Records with top-2 CShM separation $\Delta S < 1.0$. Expected behavior: shape competition is encoded explicitly in the shape field or L1 ShapeID rather than forced into an unqualified one-hot shape assignment.

SM4.8 Interpretation of “Full” in Supplementary Table S1

“Full” denotes that the feature is explicitly defined by the representation within the scope evaluated in this work; it does not imply that every chemical ambiguity in an experimental structure can be resolved from the representation alone. For CoordRep, molecular records cover mononuclear, multinuclear, bridging, and haptic/ π coordination, while periodic records retain translation-resolved local first spheres. The molecular L0–L3 hierarchy and periodic MID/SID comparison axis answer related but distinct identity questions. Source-dependent uncertainties such as unresolved disorder, protonation, or ambiguous bonding are retained as record status or provenance rather than converted into unsupported assignments.

SM4.9 Relation to downstream learning

The representation-feature comparison is distinct from downstream predictive benchmarking. Figure 4 evaluates GINE, matched CoordRep hybrids, SchNet, and ViSNet on graph-grouped property tasks, whereas Supplementary Table S1 asks which structural variables and identity operations are explicitly exposed by each

representation. A graph or Cartesian model may learn these distinctions from data, while CoordRep makes selected coordination variables directly addressable for curation, validation, retrieval, and controlled field ablation. The two roles are therefore complementary rather than mutually exclusive.

Supplementary Methods 5. Continuous geometry and CShM analyses

This section describes how established Continuous Shape Measures (CShM) are embedded into CoordRep as continuous, boundary-aware geometry fields. These analyses support the continuous-geometry fields in Figure 1, the coordination-number-specific CShM analyses in Figure 3, and the geometry-state retrieval analyses associated with the current-CSD molecular audit. CoordRep does not introduce a new geometry descriptor; rather, it places established CShM coordinates inside a canonical, fielded coordination record that also contains metal context, ligand dictionaries, stereochemical relations, validation status, and CoordRep-ID keys.

SM 5.1 Purpose and scope

Continuous Shape Measures (CShMs) are used in CoordRep to describe the geometry of a resolved first coordination sphere relative to ideal reference polyhedra. CoordRep does not introduce a new shape descriptor; rather, it incorporates established CShM coordinates into a canonical, field-addressable coordination record together with metal context, donor relations, ligand identity, and derived identity keys.

In the present work, CShM values are used for three purposes: (i) to encode graded local geometry in the CoordRep shape field; (ii) to construct the CN = 4–6 geometry atlases shown in Figure 3; and (iii) to identify structures close to competing reference shapes through the separation between the best and second-best CShM scores. The implemented reference library and its analysis scope are summarized in Supplementary Table S3.

All CShM analyses in Figure 3 describe static structural snapshots. A location in CShM space reports proximity to ideal reference geometries and is not used by itself to assign a dynamical pathway or interconversion mechanism.

SM5.2 Molecular cohort and geometry eligibility

The molecular geometry analysis uses the fixed 48,057-record tmQM/GFN2-xTB cohort employed throughout the molecular property benchmarks. For each record, the first coordination sphere was reconstructed from the retained metal and donor coordinates and evaluated against the reference polyhedra defined for its coordination number.

Valid CShM values were obtained for 35,485 records. Of these, 33,863 belong to the CN = 4–6 reference domains used in main-text Figure 3:

CN = 4: 16,682 records;

CN = 5: 6,356 records;

CN = 6: 10,825 records.

The remaining supported records belong to coordination numbers outside the three main-text atlas domains and are not included in Figure 3A–C.

SM5.3 CShM calculation and donor-to-reference assignment

For an observed coordination environment with donor coordinates q_i and a reference polyhedron P_0 with vertices p_i , the Continuous Shape Measure is defined as:

$$S(P_0) = 100 \times \min_{R, \sigma} \frac{\sum_{i=1}^N \|q_i - R_{p_{\sigma(i)}}\|^2}{\sum_{i=1}^N \|q_i - \bar{q}\|^2}$$

Here N is the coordination number, R_i is the optimal rotation, σ is the donor-to-reference-vertex assignment, and $\bar{q}$ is the centroid of the observed donor coordinates. Lower values indicate closer agreement with the reference polyhedron, with $S = 0$ corresponding to an ideal match.

For the CN = 4–6 analyses reported in Figure 3, donor-to-reference assignments were evaluated exactly in coordinate space. For each candidate assignment, rigid alignment was optimized before the minimum CShM score was selected. Coordinates were processed in double precision, and shape ranking and boundary assignment were performed before numerical rounding.

This procedure makes the reported CShM values independent of the initial donor indexing and global rigid orientation of a fixed coordination snapshot.

SM5.4 Reference shapes and nearest-reference assignment

Figure 3 compares two principal reference shapes for each coordination number:

CN = 4: square planar (*SP*) and tetrahedral (*T_d*);

CN = 5: square pyramidal (*SPY*) and trigonal bipyramidal (*TBP*);

CN = 6: octahedral (*O_h*) and trigonal prismatic (*TPr*).

For each record, the lower CShM score defines the nearest implemented reference shape. The resulting nearest-reference assignments are:

CN = 4: 13,011 *SP* and 3,671 *T_d*;

CN = 5: 5,319 *SPY* and 1,037 *TBP*;

CN = 6: 5,580 *O_h* and 5,245 *TPr*.

These assignments provide convenient discrete labels, whereas the paired CShM coordinates retain the continuous displacement of each structure from both competing references.

Coordination-number-specific indices such as τ_4 and τ_5 provide complementary scalar summaries. In CoordRep, the paired CShM values are retained because they report proximity to both reference geometries rather than reducing each structure to a single scalar or nearest-reference label.

SM5.5 Boundary definition and threshold sensitivity

For each record, the two lowest CShM values are denoted as:

$$S_{best} = \min_j S(P_j)$$

$$S_{second} = \min_{k \neq j^*} S(P_k)$$

Where j^* is the index of the best-matching reference shape.

The top-2 separation is:

$$\Delta S = S_{second} - S_{best}$$

A record is classified as near a shape boundary when

$$\Delta S < 1$$

Importantly, ΔS is calculated from the full-precision CShM values before rounding or serialization.

Using this criterion, the main-text cohort contains:

CN = 4: 39/16,682 boundary records (0.23%);

CN = 5: 693/6,356 (10.90%);

CN = 6: 294/10,825 (2.72%).

Overall, 1,026 of 33,863 CN = 4–6 records (3.03%) satisfy the boundary criterion. Thus, the CN = 5 SPY/TBP domain shows the largest relative population near the implemented shape boundary.

The dependence of these fractions on the numerical ΔS cutoff was evaluated across a broader threshold range using the same frozen cohort (Supplementary Figure S1). This sensitivity analysis is used to assess whether the relative boundary enrichment is specific to the default $\Delta S = 1$ criterion.

SM5.6 CoordRep shape-field encoding

The CoordRep geometry field retains both the nearest-reference identity and continuous CShM information. The full-precision values are used for geometry analysis and boundary assignment, whereas serialized records may use the numerical precision defined by the CoordRep grammar.

The geometry-resolved L0 record therefore preserves small differences between structural snapshots. Coarser identity levels subsequently reduce sensitivity to such differences: L1 uses boundary-aware shape/ ΔS bins, whereas L2 retains the best-shape label without the exact continuous geometry. This separation allows continuous structural variation to remain available without requiring every comparison to operate at exact-snapshot resolution.

The same explicit shape/CShM field is independently evaluated in the graph-identical property benchmark described in Figure 4D, where its contribution is measured through a matched field-hidden versus field-visible comparison.

SM5.7 Reproducibility and source data

All Figure 3 geometry analyses were regenerated from frozen source tables containing, for each retained record:

record identifier;

coordination number;

metal identity;
donor identity or donor-element summary;
CShM score for each implemented reference;
best-shape label;
 S_{best} ;
 S_{second} ;
 ΔS ;
boundary flag.

The corresponding Figure 3 and Supplementary Figure S1 source tables are included in the reproducibility package. CSD-derived analyses elsewhere in the manuscript distribute only permitted derived annotations, refcodes, identity keys, and aggregate summaries; raw licensed CSD coordinates are not redistributed.

SM5.8 Interpretation of continuous geometry fields

CShM fields are interpreted throughout this work according to three principles. First, a low CShM score indicates proximity to an ideal reference geometry. Second, a small ΔS indicates competition between the implemented reference assignments. Third, both quantities describe the geometry of a static structural snapshot.

Accordingly, continuous CShM distributions can be used to compare graded structural distortion and to construct machine-readable geometry fields, while mechanistic or dynamical interpretation requires evidence beyond the CShM coordinates themselves.

Supplementary Methods 6. Graph, GNN, and 3D equivariant baselines

SM6.1 Revised Figure 4 property-benchmark design

This section documents the frozen common-cohort property benchmark used in main-text Figure 4. The benchmark compares attributed GINE, full and field-hidden CoordRep hybrids, SchNet, and ViSNet on identical graph-grouped outer folds and three fixed random seeds. The benchmark separates two chemically distinct questions. Matched CoordRep ablations quantify the contribution of explicit donor-relation and

shape/CShM fields, whereas SchNet and ViSNet establish the performance context provided by full Cartesian coordinates on the same cohort and folds. Instead, equal-architecture CoordRep ablations test the incremental information carried by explicit donor-relation and shape/CShM fields, whereas the Cartesian models provide strong coordinate-based comparators under a common cohort and split policy.

The frozen benchmark contains 48,057 tmQM-derived records grouped into 46,212 attributed-2D-graph clusters. All models use the same five graph-disjoint folds; predictions from seeds 11, 22, and 33 are averaged recordwise before computing any metric or pair difference. Thus SchNet and ViSNet each comprise 15 completed trainings (five folds by three seeds; 288,342 raw out-of-fold prediction rows per architecture). The analysis/reporting set was locked before the complete Cartesian runs and before aggregate endpoint postprocessing or inspection.

Five reporting models are compared. GINE uses the chemically attributed 2D graph. CoordRep full combines the graph with record fields; its relation-hidden and shape/CShM-hidden variants use the same architecture and cohort, making these the direct field ablations. SchNet and ViSNet receive atomic numbers and optimized Cartesian coordinates. The Cartesian models omit the bond-order and formal/total-charge attributes available to the attributed graph, so comparisons across representation families are common-cohort/common-split predictive comparisons rather than equal-input ablations or a uniformly retuned leaderboard.

SM6.2 SchNet and modern ViSNet coordinate baselines

SchNet used four interaction blocks, 128 hidden channels, 32 Gaussian radial functions, a 5.0 Å cutoff, batch size 48, learning rate 1×10^{-3} , and 301,827 trainable parameters. ViSNet used the official PyTorch Geometric vector–scalar representation trunk with four layers, eight attention heads, 64 hidden channels, 32 radial basis functions, $l_{\text{max}} = 1$, a 5.0 Å cutoff, batch size 32, learning rate 1×10^{-4} , and 282,882 trainable parameters. Both used weight decay 1×10^{-5} , at most 60 epochs, early-stopping patience 6, and bfloat16 training.

The radius-graph neighbor cap was 96. A complete cohort census found a maximum of 81 neighbors within 5.0 Å (counting the central atom conservatively), so

no radius neighborhood was truncated. Scalar outputs passed numerical audits for proper rotation, reflection, translation, and atom permutation. The ViSNet source paper uses SE(3) terminology for its vector–scalar message-passing architecture; the reflection audit here establishes the reported scalar output invariance for the implemented predictor and should not be read as a separate formal proof for every internal feature.

The attributed GINE and CoordRep hybrids used four residual GINEConv blocks. The GINE comparator used hidden dimension 133, whereas the hybrid used hidden dimension 128 and concatenated the pooled graph embedding with a 128-dimensional embedding of train-fitted standardized CoordRep fields. Each graph block used a two-layer multilayer perceptron, batch normalization, SiLU activation, trainable ϵ , residual addition, and 0.10 dropout; graph readout used global mean pooling. The full and field-hidden hybrids preserved the same input dimension and were exactly parameter matched within each fold (337,414–337,542 trainable parameters, depending on the fold-fitted field vocabulary); the GINE comparator contained 333,836 trainable parameters. Training used AdamW with a learning rate of 1×10^{-3} , weight decay of 1×10^{-5} , batch size 256, a maximum of 40 epochs, validation patience of eight epochs, gradient clipping at 5.0, and seeds 11, 22, and 33. Target normalization and all field preprocessing were fitted on the outer-training partition only.

SM6.3 Matched-pair cohorts and estimands

Panel C uses the frozen automatic34 cis/trans cohort: 34 matched pairs sharing the attributed 2D graph. The primary error is the mean absolute error of the signed cis-minus-trans dipole difference, with direction concordance counting ties as one half. The automatic34 cohort was defined from shared attributed-graph identity and donor-relation criteria before target columns and model predictions were accessed; it therefore represents a target-blind, chemically defined pair set.

For the different-shape analysis, a pair is included only when both endpoints have nonempty best-shape assignments after trimming and the two labels differ. This corrected clean cohort contains 110 pairs in 72 attributed-graph clusters. Panel D reports the descriptive material-gap subset with observed $|\Delta(\text{HOMO} - \text{LUMO gap})| >$

0.10 eV: 44 pairs in 31 graph clusters (clean44). The observed-target threshold makes clean44 a target-enriched descriptive stress test defined independently of model predictions, not a confirmatory estimate of population-wide performance.

Overall MAE is record-weighted after three-seed averaging. Pairwise uncertainty uses 20,000-replicate ordinary nonparametric bootstrap resampling of the frozen attributed-graph cluster, retaining the original record or pair weighting. Paired sign-flip tests use 200,000 randomizations. Percentile intervals are conditional on the frozen three-seed ensemble, descriptive tail probabilities are not null-calibrated p values, and no multiplicity adjustment was applied.

SM6.4 Interpretation boundary

The equal-architecture CoordRep ablations isolate record-field contributions. In automatic34, restoring relation fields lowers dipole pair-difference MAE from 5.218489 to 1.327899 D (74.55%) and raises direction concordance from 79.4% to 94.1%; SchNet and ViSNet remain stronger in absolute dipole prediction. In clean44, restoring shape/CShM fields lowers gap-difference MAE from 0.359824 to 0.245444 eV (31.79%) and raises direction concordance from 67.0% to 84.1%. These results support state-discriminating, complementary information in CoordRep fields; Together, these matched contrasts identify donor relations and shape/CShM fields as measurable state-discriminating variables, while the Cartesian baselines define the task-dependent performance context.

CoordRep does not calculate redox thermodynamics, vibronic coupling, covalency, spin-state energetics, or a unique electronic Hamiltonian; these quantities require independent electronic-structure calculations or experimentally validated predictive models.

SM6.5 Frozen source data and ViSNet citation

The authoritative public Figure 4 source tables are archived under figures/Figure4/, the SchNet and ViSNet configurations and audit reports under models/E3_baselines/, and the frozen protocols under protocols/. Licensed row-level CSD structures and raw coordinates are not redistributed. Required files are overall_metrics.csv,

cis_trans_metrics.csv, cis_trans_contrasts.csv, cis_trans_pair_predictions.csv, different_shape_pair_metrics.csv, different_shape_pair_contrasts.csv, different_shape_pair_predictions.csv, clean44_gap_pair_predictions.csv, AUDIT.json, and SHA256SUMS.txt. The v3 protocol supersedes only the earlier different_shape and corrected60 postprocessing labels, metrics, figures, and prose; no model was retrained and no out-of-fold prediction changed.

Supplementary Methods 7. April 2025 CSD molecular and canonical periodic audits

SM7.1 Scope, denominators, and molecular analysis tiers

All 1,371,757 entries in the locally installed April 2025 Cambridge Structural Database (CSD) were scanned. Entries containing three-dimensional coordinates and at least one element in the predefined d-block target domain formed the operational target set of 602,116 entries. This set comprises 253,522 nonpolymeric entries with a single in-domain metal center, 233,622 nonpolymeric entries with multiple in-domain centers or an external-metal relation, and 114,972 polymeric entries.

Two more restrictive molecular subsets are used for analyses that require additional chemical resolution. The strict mononuclear η^1 core contains 101,878 records satisfying coordinate, domain, disorder/polymer, donor-count, corrected-CShM, syntax, and mononuclear record gates. A connectivity-supported molecular tier of 84,453 records additionally satisfies the operational CSD-bond-topology ligand route and is used for molecular family and identity analyses. These subsets are analysis tiers within the 602,116-entry target corpus and should not be added to the mutually exclusive full-CSD partitions above.

SM7.2 Expanded molecular and periodic record construction

Each native CSD structure was converted into a typed metal-site–donor-group incidence record. Every in-domain d-block atom defines a metal-site node. An atom-resolved η^1 donor is retained as a one-atom donor group. A connected 2–12-atom component induced by native π /delocalized metal–donor bonding is represented as a

single π -fragment site with explicit atom membership and hapticity. Larger collective- π components are retained as candidate atom sets with undefined η and an ambiguity flag rather than assigned a forced hapticity.

A donor group shared by multiple metal centers is stored once together with its complete target-metal set and bridge degree. One typed incidence is emitted for every native metal–donor-member contact. Direct metal–metal bonds, donor-group overlaps, and relations to metals outside the target domain are retained separately. For polymeric structures, each native incidence or metal relation additionally carries an integer lattice-translation label τ obtained from the source fractional-coordinate lift.

Across the 602,116-entry target corpus, 147,348 entries contain at least one donor group shared by in-domain metal centers, 102,241 contain at least one confirmed haptic/ π site, and 94,452 contain at least one resolved nonzero lattice-translation edge. These categories are nonexclusive. Thirty-four entries contain at least one oversized collective- π candidate for which a definite hapticity is intentionally not assigned.

SM7.3 Source-fidelity validation and structural-record retention

An independent validator reread every native CSD object and compared the source structure with the emitted CoordRep audit record. The comparison included the in-domain metal inventory, metal–donor incidences, local coordination counts, shared-donor and bridge signatures, haptic-site membership, metal–metal relations, donor-pair geometry, and periodic-translation flags. All 602,116 target entries yielded schema-valid audit records, and all reproduced the corresponding source-derived signatures without mismatch.

A stricter structural-record criterion requires a resolved local record for each retained in-domain metal site. Structural records were obtained for all in-domain metal sites across 601,402 entries (99.88% of the target entries), accounting for 1,984,062 of 1,986,198 metal centers (99.89%). Entry-level terminal outcomes comprise 293,850 resolved, 76,608 partial, 231,206 disorder-ambiguous, and 452 audit-only entries. These status labels preserve limitations in the source model rather than silently excluding the corresponding entry.

For the 114,972 polymeric entries, 114,916 retain complete translation information

for the implemented periodic record construction (99.95%). Missing geometry, unresolved translations, disorder, or ambiguous collective donor assignments reduce the retained record resolution but do not invalidate the schema-level audit record.

SM7.4 Multi-resolution molecular identity and family analysis

Molecular identity was evaluated at four nested resolutions. L0 (CoordRep-State) is the complete canonical geometry-resolved record and retains the exact CShM values, donor-site map, oxidation state, d-electron count, metal identity, coordination number, and ligand payload. L1 (ShapeID) retains metal identity, oxidation state, coordination number, ligand identities, boundary-aware shape/ ΔS bins, and stereochemical donor relations. L2 (TopoID) retains metal identity, coordination number, the best-shape label, and ligand identities. L3 groups records by metal identity, coordination number, and ligand-payload family; formal ConnID assignment additionally requires SMILES-resolved provenance. Equality is nested such that an L0 match necessarily implies L1, L2, and L3 matches.

Within the 84,453-record connectivity-supported molecular tier, 3,491 CSD refcode families contain at least two records, comprising 9,056 records in total. Mean pairwise agreement within these families is 11.2% at L0, 83.5% at L1, 96.6% at L2, and 97.4% at L3. Thus, L0 remains sensitive to the experimentally resolved geometry state, whereas progressively coarser levels recover shared shape and connectivity-level family identity.

The five-record XEYVEC $[\text{CuCl}_4]^{2-}$ family provides a representative hierarchy test. The five crystallographic determinations resolve into five L0 states, two L1 identities, two L2 identities, and one L3 family. The two L1/L2 groups correspond to the T_d -leaning and SP -leaning regions discussed in the main text.

For the separate common attributed-graph comparison intersection, the atom- and bond-attributed Weisfeiler–Lehman key and the relation-augmented key were evaluated on the same frozen records. Adding CoordRep donor-pair relations increases same-key pair precision from 0.828 to 0.889. This comparison is used only to quantify the additional discrimination supplied by explicit coordination relations and is separate from the full 84,453-record molecular hierarchy.

SM7.5 Canonical periodic quotient and the XUMMID regression

Periodic atoms are first grouped by wrapped fractional-coordinate equivalence. Each metal–donor edge retains both the wrapped donor orbit and the relative integer lattice-translation vector τ . The physical metal–donor vector is reconstructed in Cartesian space before local geometry and CShM evaluation.

Canonical metric identity is constructed from chemical site attributes, donor-orbit equality partitions, and the donor-vector Gram matrix. Orientation-sensitive stereo identity additionally includes signed donor triple products. After fixed numerical quantization, admissible donor permutations are enumerated for $CN = 2\text{--}6$ and the lexicographically minimal complete state record is selected. Translation-labelled quotient edges remain explicitly available in the auditable record.

XUMMID provides a regression test for translation-resolved first-sphere reconstruction. The Cu site contains four Cu–N contacts of 2.062 Å and two Cu–Cl contacts of 2.696 and 2.768 Å. The two chloride contacts correspond to the same wrapped donor orbit but to different relative lattice translations. Retaining both incidences reconstructs the physical CuN_4Cl_2 first sphere with $CN = 6$ and $S(\text{Oh}) = 1.8748$. Collapsing the two contacts solely by wrapped donor identity instead produces an artificial five-coordinate CuN_4Cl state. XUMMID is used as an implementation regression and illustration of the periodic quotient rather than as an independent validation set.

SM7.6 CSD MOF Collection census and MID/SID resolution

The periodic analysis used the public 15,906-entry CSD MOF Collection in its supplied *P1* representation. Canonical processing completed for 15,905 entries, of which 10,948 emitted at least one local metal-site state. In total, 172,332 translation-resolved local states were generated. Of these, 74,354 states (43.15%) contain at least one nonzero translation-labelled edge, and 9,805 of the 10,948 state-bearing entries contain at least one such state.

The parity-free metric layer resolves the 172,332 local states into 36,552 metric identities (MIDs), whereas the orientation-sensitive layer resolves 60,139 stereo

identities (SIDs). Every MID maps to either one or two SIDs: 12,965 MIDs remain unsplit and 23,587 split into exactly two SIDs. The split counts are 0/1,041 for CN = 2, 559/812 for CN = 3 (68.8%), 5,722/9,209 for CN = 4 (62.1%), 7,918/9,915 for CN = 5 (79.9%), and 9,388/15,575 for CN = 6 (60.3%).

A stricter same-orbit/different-translation multiedge motif occurs at 108 sites in 28 entries. An explicit-image internal-consistency witness could be reconstructed for 169,900 of the 172,332 emitted states (98.59%). These quantities characterize translation-resolved local first spheres and do not constitute whole-framework net identities.

SM7.7 Association between local stereo pairing and crystallographic symmetry

MID-to-SID pairing was derived entirely from the supplied P1 coordinates before the independently stored original-space-group annotation was joined. A within-entry coexistence event was defined when one MID occurred with both of its opposite-orientation SIDs in the same entry.

Such pairing occurs in 7,721 of 9,189 state-bearing entries not flagged as Sohncke (84.02%) and in none of the 1,759 Sohncke entries. The association is consistent with the presence of orientation-reversing crystallographic operations in non-Sohncke space groups and the restriction of Sohncke space groups to orientation-preserving operations. The analysis is therefore interpreted as a local orientation-sensitive symmetry signature; MID/SID pairing is not used as an assignment of whole-framework chirality, enantiopurity, or a continuous chirality magnitude.

SM7.8 Periodic canonical-invariance audit

Periodic canonicity was challenged by compound crystallographic re-expression rather than by isolated atom permutations alone. The primary audit comprised 400 family-disjoint entries containing 5,920 local metal sites. For each entry, simultaneous atom/donor permutations, global fractional-origin translations, independent periodic-image and quotient-orbit relabellings, and determinant-positive unimodular lattice-basis changes were applied.

All 400/400 transformed entries retained their complete entry-state multisets, and

all 5,920 local MIDs and SIDs were unchanged under the orientation-preserving transformations. A separate set of 50 determinant-negative basis comparisons retained MID in 50/50 cases; SID was not required to remain invariant under orientation reversal because its defining signed triple products intentionally distinguish opposite local orientations.

These tests establish invariance to the crystallographic descriptions explicitly included in the canonicalization domain. The full-CSD and CSD MOF analyses therefore report typed, source-faithful local structural records and canonical local identities, while inheriting the coordination assignments, disorder model, protonation state, and structural resolution of the source crystallographic record.

Supplementary Methods 8. Reproducibility package and file manifest

SM8.1 Active evidence package

The active evidence package contains serializer examples; the attachment-aware canonicalizer; the general-invariance and legal-orbit challenges; CoordRep-ID and CShM analyses; frozen graph-grouped property splits; GINE, CoordRep-hybrid, complete-radius SchNet, and ViSNet configurations and predictions; the leakage-controlled composite-versus-factorized tokenizer ablation; the frozen-checkpoint field-block order stress test; the conditional multidentate donor-field evaluation; current-CSD molecular and attributed-WL outputs; expanded molecular site-topology outputs; canonical periodic outputs; figure-ready source tables; analysis protocols; and SHA-256 checksums.

SM8.2 Property comparison source

The authoritative local source is `revision_experiments/results/coordstatepairs_cartesian_complete_radius_postprocessing_v3_clean_shapes/`. The postprocessor audit status is PASS. Corrected complete-radius SchNet and ViSNet each contribute 288,342 raw out-of-fold rows (five folds, three seeds, two targets).

SM8.3 Excluded developmental analyses

The former inverse-design and ligand-recommendation framing is not part of the

active evidence. The retained donor-set analyses are bounded field-recovery diagnostics: they test whether a recorded donor field can be recovered from controlled record contexts. They do not generate ligands, establish donor compatibility, predict thermodynamic stability or synthetic accessibility, or validate a proposed molecular structure. Historical cross-domain retrieval and development-only files are excluded from the active submission evidence.

SM8.4 Data access

Public aggregate source tables, protocols, figures, and checksums are archived in release/jacs-revision-20260823 on the jacs-revision branch. Licensed row-level CSD structures, coordinates, and structure mappings remain local and are not redistributed.

SM8.5 Source data for sequence-interface, canonicalization, donor-context, and attributed-WL analyses

Supplementary Table S7 is generated from models/tokenizer_factorization/. Its authoritative files are dataset_audit.json, split_ids.csv, metrics_by_seed.csv, metrics_summary.csv, paired_bootstrap.csv, test_predictions.csv, run_metadata.json, vocab_composite.json, vocab_factorized.json, and SHA256SUMS.txt.

Supplementary Table S8 is generated from models/field_order_production/. Its authoritative files are masked_field_summary.csv, masked_field_paired_bootstrap.csv, masked_field_per_prediction.csv, semantic_decoy_summary.csv, semantic_decoy_paired_bootstrap.csv, semantic_decoy_per_pair.csv, token_multiset_audit.csv, run_metadata.json, source_manifest.json, and SHA256SUMS.txt.

Figure 2 and Supplementary Table S2A are generated from canonicalization/general_invariance/ and canonicalization/legal_orbit_challenge/. The first directory contains the locked cohort manifest, per-arm variant summary, mismatch ledger, run metadata, implementation provenance, and checksums. The second contains the preregistered protocol, complete-screen manifest, locked challenge-cohort source table, aggregate summary, run log, and checksums. Supplementary Table S2B is generated from canonicalization/expanded_record_regressions/. Supplementary Table S14A is generated from models/multidentate_context/, and Supplementary Table S14B from models/multidentate_conditional/. No Supplementary Figure S3 is part of the active submission evidence.

The attributed-WL control underlying Supplementary Table S10 is archived under

identities/attributed_wl_current_csd/. Public aggregate files are metrics.csv, family_metrics.csv, coverage_failures.csv, permutation_audit.csv, SUMMARY.json, and SHA256SUMS.txt. Licensed refcode-level graph and coordinate tables remain local and are not redistributed.

Supplementary Results

Supplementary Results 1. Modern common-cohort property comparison

The full common-cohort results establish a target-dependent boundary. CoordRep hybrids are competitive for HOMO–LUMO gap, whereas complete-radius SchNet and ViSNet are substantially stronger for dipole. The equal-architecture field ablations, rather than a universal model rank, provide the direct test of CoordRep-specific information.

Supplementary Results 2. Graph-identical cis/trans pairs

On automatic³⁴, restoring donor relations lowers the matched hybrid dipole pair-difference MAE from 5.2185 to 1.3279 D (74.55%) and raises direction concordance from 0.7941 to 0.9412. SchNet and ViSNet give 0.8002 and 0.4815 D, respectively, with concordance 1.000. This cohort was selected without reading targets or predictions but was discovered during corpus exploration.

Among the 31 relation-resolved cis/trans pairs shown in Figure 4B, 30 have a larger computed dipole for the cis member, and the median absolute cis–trans separation is 6.0746 D. These pairs have identical attributed two-dimensional graphs and differ in the recorded donor-relation pattern.

Supplementary Results 3. Corrected different-shape gap stress test

Requiring nonmissing but different best-shape assignments at both endpoints leaves 110 pairs in 72 graph clusters. The target-enriched $|\Delta_{\text{gap}}| > 0.10$ eV subset contains 44 pairs in 31 clusters. GINE, the shape-hidden hybrid, full CoordRep hybrid, SchNet, and ViSNet give pair-difference MAEs of 0.5356, 0.3598, 0.2454, 0.3747, and 0.3907 eV. Restoring shape/CShM lowers the matched hybrid error by 31.79% and

raises concordance from 0.6705 to 0.8409. Because selection uses the observed target difference, this is a descriptive stress test. The 44-pair gap cohort was conditioned on an observed $|\Delta_{\text{gap}}| > 0.10$ eV and is therefore a focused descriptive contrast rather than an unconditional estimate over all same-graph pairs.

Supplementary Results 4. Molecular and periodic database applications

The April 2025 CSD audit covers 602,116 three-dimensional d-block entries. All target entries yielded schema-valid audit records, while structural records were retained for 1,984,062 of 1,986,198 in-domain metal sites (99.89%) across 601,402 entries. Within the 84,453-record connectivity-supported molecular tier, 3,491 multi-record CSD refcode families comprising 9,056 records were used to evaluate the L0–L3 identity hierarchy, which progressively links the finest serialized coordination states at the stated precision to broader coordination families. In the public CSD MOF Collection, canonical processing completed for 15,905 of 15,906 entries and generated 172,332 translation-resolved local states, resolving 36,552 metric identities (MIDs) and 60,139 orientation-sensitive stereo identities (SIDs). Together, these analyses establish database-scale molecular family linking, explicit treatment of multinuclear, haptic, and periodic coordination modes, and canonical local-state recording across molecular and periodic coordination environments.

Supplementary Tables

Supplementary Table S1. Operation coverage of representative molecular and coordination-chemistry representation families.

Representation family	Canonical record identity	Coordination stereochemistry	Continuous / boundary geometry	Family linking across geometry states	Validation-ready record	Multi-resolution identity
Conventional molecular strings ^{4,5,15,25}	Partial: canonical strings exist for many covalent molecules, but coordination bonding is incomplete	Partial: conventional stereochemistry is supported, but donor-site relation maps are not explicit	Not supported	Partial: connectivity-level linking is possible for simple cases	Partial: syntax can be parsed, but coordination-field consistency is absent	Not supported
TMC structure-to-SMILES workflows ²⁸	Partial: improves TMC coordinate-to-string conversion	Partial: depends on the supported stereochemical-perception procedure	Not supported	Partial: connectivity-level linking is possible after conversion	Partial: output is parseable, but complete coordination-record validation is not provided	Not supported
TMC descriptor / featurization schemes	Not supported: descriptors are not canonical records	Partial: may include stereochemical or geometric features	Partial: may include geometry descriptors	Not supported	Not supported	Not supported
Stereoisomer / conformer workflows	Partial: may define canonical objects within supported enumerations	Full for supported classes	Partial: conformer geometry may be generated, but boundary geometry is not the identity object	Not supported	Partial: generated structures can be checked, but record grammar is not central	Not supported
m-SMILES / MetalloGen-style notations ²⁹	Partial: coordination-aware string for generation	Partial	Partial: geometry may be implied or generated	Partial: related strings may be compared,	Partial: string syntax exists, but not full CoordRep-style validation	Not supported

Representation family	Canonical record identity	Coordination stereochemistry	Continuous / boundary geometry	Family linking across geometry states	Validation-ready record	Multi-resolution identity
				but no identity ladder is defined		
T-REX trans-pair notation ³²	Full within the supported domain	Full: trans-pair relations are explicitly encoded	Partial: geometry-resolved notation, but without complete CShM boundary fields	Partial: related records can be compared, but no L0–L3 hierarchy is defined	Partial: the notation can be parsed, but its validation scope differs	Not supported
CShM-only / coordinate-based geometry ¹⁷⁻²¹	Not supported	Partial: some relation information can be inferred from coordinates	Full: continuous shape values are the primary object	Not supported: no ligand or connectivity identity is defined	Not supported	Not supported
Attributed WL graph key ²⁴	Full for deterministic graph identity within the chosen attributed graph	Full when the relevant stereochemical relations are supplied as graph attributes; otherwise partial	Not supported for continuous or boundary-resolved geometry	Full for connectivity-family linking	Partial: graph/key construction is parseable and auditable but is not a complete coordination-record validator	Not supported
CoordRep	Full	Full	Full	Full	Full	Full

Supplementary Table S2A. End-to-end mononuclear canonicalization, legal-orbit, and stereochemical-specificity tests supporting Figure 2.

Test	Perturbation/control	N	CoordRep result		Control or interpretation	
General invariance	Proper rigid motion	$1,000 \times 100$	100,000/100,000 matches; collapsed	1,000/1,000	Cartesian ordered 0/1,000; distance 1,000/1,000; source order 997/1,000	
General invariance	Atom/connection-table reindexing	$1,000 \times 100$	100,000/100,000; 1,000/1,000		Cartesian ordered 0/1,000; distance 0/1,000; source order 50/1,000	
General invariance	Combined	$1,000 \times 100$	100,000/100,000; 1,000/1,000		Cartesian ordered 0/1,000; distance 0/1,000; source order 50/1,000	
Legal-orbit screen	Eligible graphs with >1 legal candidate	5,173/6,427	Challenge identified	population	No outcome-dependent selection	
Signature-only ablation	Atom reindexing	$1,000 \times 100$	64,834/100,000; 380/1,000 collapsed		Exact enumeration omitted	
Full exact search	Same variants	$1,000 \times 100$	100,000/100,000; 1,000/1,000 collapsed		100,000 round trips; stable candidates; zero errors	
Attachment specificity	Adjacent vs opposite attachment sets	1 pair	Distinct records; false merges	zero	Nonautomorphic negative control	
Missing metadata	Repeated payload without attachment key	1 constructed case	Rejected		Fail-closed control	
fac/mer specificity	Idealized fac/mer-[Co(NH ₃) ₃ Cl ₃]	2 models	Distinct CoordRep-States		Same metal/CN/ligand/ideal-Oh fields; different trans inventories	

Supplementary Table S2B. Synthetic exact-tie and expanded-record canonicalization regressions.

Test	Perturbation/control	N	Expected result	Observed
CN = 4 exact ties	Within-class permutations	4/4	One state	Pass

Test		Perturbation/control	N	Expected result	Observed
CN = 6 exact ties		Within-class permutations	8/8	One state	Pass
Cu ₄ cycle		All metal orderings	24/24	One state	Pass
Topology control	negative	Cu ₄ cycle vs path	1 pair	Distinct	Pass
Multimetal typed records		55 templates × variants	100 5,500	Unchanged	5,500/5,500
Haptic/ π typed records		33 templates × variants	100 3,300	Unchanged	3,300/3,300

These curated programmatic templates are regression tests for the typed record abstraction and are not verified CSD structures or prevalence samples.

Supplementary Table S3. Reference polyhedra used for CShM calculations and their analysis scope.

CN	Shape Label	Ideal Polyhedral Geometry	Point Group	Analysis Scope
2	<i>L</i>	Linear	$D_{\infty h}$	Expanded reference library
3	<i>TP</i>	Trigonal planar	D_{3h}	Expanded reference library
3	<i>TY</i>	T-shaped	C_{2v}	Expanded reference library
4	<i>SP</i>	Square planar	D_{4h}	Main-text CN = 4 atlas
4	<i>Td</i>	Tetrahedral	T_d	Main-text CN = 4 atlas
5	<i>SPY</i>	Square pyramidal	C_{4v}	Main-text CN = 5 atlas
5	<i>TBPY</i>	Trigonal bipyramidal	D_{3h}	Main-text CN = 5 atlas
6	<i>Oh</i>	Octahedral	O_h	Main-text CN = 6 atlas
6	<i>TPr</i>	Trigonal prismatic	D_{3h}	Main-text CN = 6 atlas
7	<i>PBPY</i>	Pentagonal bipyramidal	D_{5h}	Expanded reference library
7	<i>COC</i>	Capped octahedral	C_{3v}	Expanded reference library
8	<i>SAPR</i>	Square antiprismatic	D_{4d}	Expanded reference library
8	<i>CU</i>	Cubic	O_h	Expanded reference library
9	<i>TCTP</i>	Tricapped trigonal prismatic	D_{3h}	Expanded reference library

Supplementary Table S4. Overall property-prediction MAEs on the common 48,057-record graph-grouped cohort.

Target	Model	N records	MAE	95% CI
HOMO–LUMO gap (eV)	GINE	48057	0.242569	0.240285–0.244897
HOMO–LUMO gap (eV)	CoordRep shape/CShM-hidden	48057	0.232330	0.230129–0.234580
HOMO–LUMO gap (eV)	CoordRep full	48057	0.230744	0.228540–0.232960
HOMO–LUMO gap (eV)	SchNet	48057	0.249534	0.247080–0.252010
HOMO–LUMO gap (eV)	ViSNet	48057	0.253271	0.250719–0.255787
Dipole moment (D)	GINE	48057	1.561064	1.543979–1.578143
Dipole moment (D)	CoordRep shape/CShM-hidden	48057	1.510774	1.494916–1.527211
Dipole moment (D)	CoordRep full	48057	1.495054	1.479169–1.511317
Dipole moment (D)	SchNet	48057	0.702303	0.693012–0.712105
Dipole moment (D)	ViSNet	48057	0.589798	0.581499–0.598720

Supplementary Table S5. Target-blind graph-identical 34-pair cis/trans dipole benchmark.

Model	Pair MAE (D)	95% CI	Direction	95% CI
GINE	5.868915	4.708608–7.055230	50.0%	50.0–50.0%
CoordRep relation-hidden	5.218489	4.141020–6.333824	79.4%	66.2–91.2%
CoordRep relation-full	1.327899	0.997259–1.684453	94.1%	86.8–100.0%
SchNet	0.800163	0.568940–1.064443	100.0%	100.0–100.0%
ViSNet	0.481468	0.351697–0.617562	100.0%	100.0–100.0%

Supplementary Table S6. Focused graph-identical 44-pair shape-competition HOMO–LUMO-gap benchmark.

Model	Pair MAE (eV)	95% CI	Direction	95% CI
GINE	0.535594	0.275651–0.866358	50.0%	50.0–50.0%
CoordRep shape/CShM-hidden	0.359824	0.245116–0.484402	67.0%	47.8–83.7%
CoordRep full	0.245444	0.195132–0.288643	84.1%	70.5–95.0%
SchNet	0.374699	0.212593–0.559083	77.3%	62.8–89.7%
ViSNet	0.390732	0.244132–0.550020	77.3%	59.5–91.2%

Supplementary Table S7. Leakage-controlled composite-versus-factorized metal/CN header ablation. Values are mean $\pm$ sample standard deviation over three model seeds on the same 5,063-record ligand-grouped test set. The primary columns use matched complete-header masking. Single-field interventions are not expressible for the composite token.

Tokenization	Metal Top-1 (%)	CN Top-1 (%)	Metal+CN joint Top-1 (%)	Exact header Top-1 (%)	Metal-only Top-1 (%)	CN-only Top-1 (%)
Composite metal/CN	60.46 $\pm$ 0.15	89.53 $\pm$ 0.21	54.98 $\pm$ 0.15	54.98 $\pm$ 0.15	not expressible	not expressible
Factorized metal + CN	66.85 $\pm$ 0.18	89.75 $\pm$ 0.15	60.27 $\pm$ 0.17	58.36 $\pm$ 0.21	63.09 $\pm$ 4.19	89.70 $\pm$ 0.05

The factorized-minus-composite paired differences were +6.39 percentage points for metal recovery (95% bootstrap CI, +5.48 to +7.28), +0.22 points for CN recovery (−0.16 to +0.63), +5.29 points for joint metal+CN recovery (+4.38 to +6.19), and +3.38 points for exact-header recovery (+2.48 to +4.30). The metal-only and CN-only columns are secondary inference-time addressability diagnostics after the matched complete-header training.

Supplementary Table S8. Frozen-checkpoint inference-time field-block order stress test. The fixed-global condition uses ligands → relations → shape → metal. Record-random values are calculated after averaging scores across deterministic order seeds 11, 22, and 33. The exact post-truncation token multiset was retained in every condition.

Analysis	Evaluation size	Metric	Canonical	Fixed global	Record random
Donor-field recovery	4,000 targets; 2,000 complexes	Top-1 (%)	85.575	84.300	84.958
Structure-field recovery	1,966 records	Top-1 (%)	97.406	94.151	96.253
Strict semantic-decoy ranking	1,450 groups; 10,877 pairs	AUROC	0.7981	0.5589	0.6706
Stereo-hard semantic ranking	960 groups; 2,254 pairs	AUROC	0.9475	0.5917	0.7910

Canonical-minus-fixed and canonical-minus-record-random differences were, respectively: donor Top-1, +1.275 percentage points (95% cluster-bootstrap CI, +0.600 to +1.975) and +0.617 points (+0.167 to +1.075); structure-field Top-1, +3.255 points (+2.391 to +4.171) and +1.153 points (+0.610 to +1.695); strict AUROC, +0.2392 (+0.2302 to +0.2479) and +0.1275 (+0.1194 to +0.1355); and stereo-hard AUROC, +0.3558 (+0.3481 to +0.3627) and +0.1564 (+0.1462 to +0.1670).

Supplementary Table S9A. Release-wide molecular and periodic dataset census and audit scope.

Analysis	Quantity	Count	Interpretation
April 2025 CSD	All entries examined	1,371,757	Release-wide census denominator
Full-CSD audit	3D entries containing ≥ 1 in-domain d-block center	602,116	Frozen audit target set: Sc–Zn, Y–Cd, La, Hf–Hg
Full-CSD audit	Schema-valid audit records emitted	602,116/602,116 (100.0000%)	Every target received a typed record outcome
Full-CSD audit	Independent native-source fidelity validation	602,116/602,116 (100.0000%)	Native metal inventory and source-derived coordination signatures reproduced
Structural coverage	Entries containing ≥ 1 native structural metal-site record	601,664/602,116 (99.9249%)	Entry-level any-site structural coverage
Structural coverage	Entries with every in-domain metal site structurally recorded	601,402/602,116 (99.8814%)	Strict entry-level structural coverage
Structural coverage	Native structural in-domain metal sites	1,984,062/1,986,198 (99.8925%)	Site-level structural coverage
Full-CSD scope	Strict nonpolymeric single-center entries	253,522 (42.1052%)	Exactly one in-domain center and no external-metal relation
Full-CSD scope	Nonpolymeric multiple-center or external-metal-relation entries	233,622 (38.8002%)	Multiple in-domain centers or a typed external-metal relation
Full-CSD scope	Polymeric entries	114,972 (19.0947%)	Mutually exclusive with the preceding two classes
Chemical relations	Entries with an in-domain shared donor group	147,348 (24.4717%)	Nonexclusive bridging relation
Chemical relations	Entries with a confirmed haptic/ π site	102,241 (16.9803%)	Graph-resolved π fragment with $\eta > 1$
Chemical relations	Entries containing an oversized collective- π candidate	34	No definite hapticity was forced
Chemical relations	Candidate entries also containing a confirmed haptic/ π site	7/34	Candidate status does not negate a separate confirmed site
Periodic relations	Entries with ≥ 1 resolved nonzero translation-labelled edge	94,452 (15.6867%)	Nonexclusive periodic-edge relation
Periodic completeness	Translation-complete polymeric source graphs	114,916/114,972 (99.9513%)	The remaining 56 entries retained typed unresolved outcomes
Periodic completeness	Entries with unresolved periodic-edge evidence	9	Retained as typed unresolved outcomes
Periodic completeness	Edge-less audit-only polymeric outcomes	47	No resolved periodic edge available

Analysis	Quantity	Count	Interpretation
CSD MOF Collection	Processed entries	15,905/15,906 (99.9937%)	Canonical-v3 periodic processing
CSD MOF Collection	State-bearing entries	10,948	Processed entries emitting ≥ 1 accepted CN2–6 local state
CSD MOF Collection	Emitted local states	172,332	Periodic local metal-site states
CSD MOF Collection	States with ≥ 1 nonzero translation-labelled edge	74,354/172,332 (43.15%)	Translation-resolved periodic local states
Canonical perturbation	Family-disjoint crystallographic re-expression challenge	400/400 entries; 5,920 local sites	Complete local-state and MID/SID multisets unchanged under tested proper re-expressions

Supplementary Table S9B. Mutually exclusive terminal outcomes for the 602,116-entry target corpus.

Audit level	Entries	Fraction of the 602,116-entry 3D d-block target set
Resolved	293,850	48.8029%
Partial	76,608	12.7231%
Disorder-ambiguous	231,206	38.3989%
Audit-only	452	0.0751%
Total	602,116	100.0000%

Supplementary Table S10. Identity-key agreement and same-key precision on the frozen attributed-WL comparison intersection.

Identity key	Within-family pair agreement	Same-key pair precision
CoordRep L0	0.218	0.946
CoordRep L1	0.708	0.898
CoordRep L2	0.766	0.850
CoordRep L3	0.824	0.814
Attributed WL	0.936	0.828
Attributed WL + cis/trans relations	0.900	0.889

These values are calculated on the frozen 2,112-record, 752-family common WL comparison intersection and are separate from the 84,453-record connectivity-supported tier and the 3,491 refcode families used for the main-text identity hierarchy.

Supplementary Table S11. CoordRep-ID hierarchy and retained information.

Layer	Identifier Name	Encoded Fields & Structural Content	Intended Scope & Primary Use
L0	CoordRep-State	Metal identity, available oxidation state and d-electron count, CN, rounded serialized CShM and shape fields, complete donor-pointer relation map, and provenance-tagged ligand payloads	Finest encoded coordination-state equality at the stated serialization precision; deterministic state tracking and same-source record comparison. Not a lossless Cartesian-coordinate identity.
L1	ShapeID	Metal identity, available oxidation state, CN, boundary-aware binned-shape token, relation-class/count signature, and sorted ligand identities	Geometry-basin grouping while retaining coarse shape competition and relation content
L2	TopoID	Metal identity, CN, best-shape label, and sorted ligand identities	Best-shape coordination-family linking without shape-bin or donor-relation detail
L3	ConnID	Metal identity, CN, sorted provenance-tagged ligand payloads, and connectivity-support status	Geometry-free coordination-family linking when all ligand payloads have connectivity support

Formal ConnID is reported only when every ligand payload has SMILES-resolved connectivity provenance. The hierarchy is nested only for records for which all four keys are defined under the same serializer and identity-key version.

Supplementary Table S12. Periodic MID/SID resolution summary.

CN	MID	Unsplit MIDs (1 SID)	Split MIDs (2 SIDs)	Stereochemical Split Fraction (%)	Total Emitted SIDs
2	1,041	1,041	0	0.0%	1,041
3	812	253	559	68.8%	1,371
4	9,209	3,487	5,722	62.1%	14,931
5	9,915	1,997	7,918	79.9%	17,833
6	15,575	6,187	9,388	60.3%	24,963
Total	36,552	12,965	23,587	64.5%	60,139

Metric identity (MID) captures labelled metric geometry through metal/site attributes, donor-vector Gram information, and radial terms. Stereo identity (SID) adds the signed scalar-triple-product tuple of the canonically ordered donor vectors. An unsplit MID has one observed SID, whereas a split MID has exactly two observed SIDs with opposite local orientation parity. “Split” does not by itself establish macroscopic framework chirality, enantiopurity, or a continuous chirality magnitude.

The total satisfies $N(\text{SID}) = N(\text{unsplit}) + 2N(\text{split}) = 12,965 + 2 \times 23,587 = 60,139$.

Supplementary Table S13. Canonical invariance, crystallographic-symmetry association of local orientation pairing, and coordinate re-expression benchmark.

A. Association of within-entry MID-to-SID pairing with crystallographic space-group class.

Space-group flag	Symmetry constraint	State-bearing entries	Entries containing one MID represented by two SIDs	MID Fraction
Not flagged as Sohncke	Improper crystallographic operations may be present	9,189	7,721	84.02%
Flagged as Sohncke	Only proper crystallographic operations	1,759	0	0.00%

B. Canonical invariance under coordinate and lattice-basis re-expression.

Transformation Category	Mathematical Transformation Scope	Target Sites Tested	MID Invariance Rate	SID Invariance Rate
Orientation-preserving (det M > 0)	Atom/donor permutations, origin shifts, quotient-orbit relabeling, and proper unimodular basis changes	5,920	5,920/5,920 (100%)	5,920/5,920 (100%)
Orientation-reversing (det M < 0)	Improper basis re-expression	50	50/50 (100%)	Not evaluated as an invariance criterion; SID is intentionally orientation-sensitive

MID is invariant under both proper and improper orthogonal re-expressions because it depends on Gram and radial terms. SID additionally contains signed triple products and is invariant only under orientation-preserving transformations. An orientation-reversing operation is therefore expected to reverse the parity tuple rather than preserve SID. The 50 determinant-negative comparisons reported here validate MID only and are not an empirical SID-inversion benchmark.

Supplementary Table S14A. Donor-site context ablation in bidentate ligands

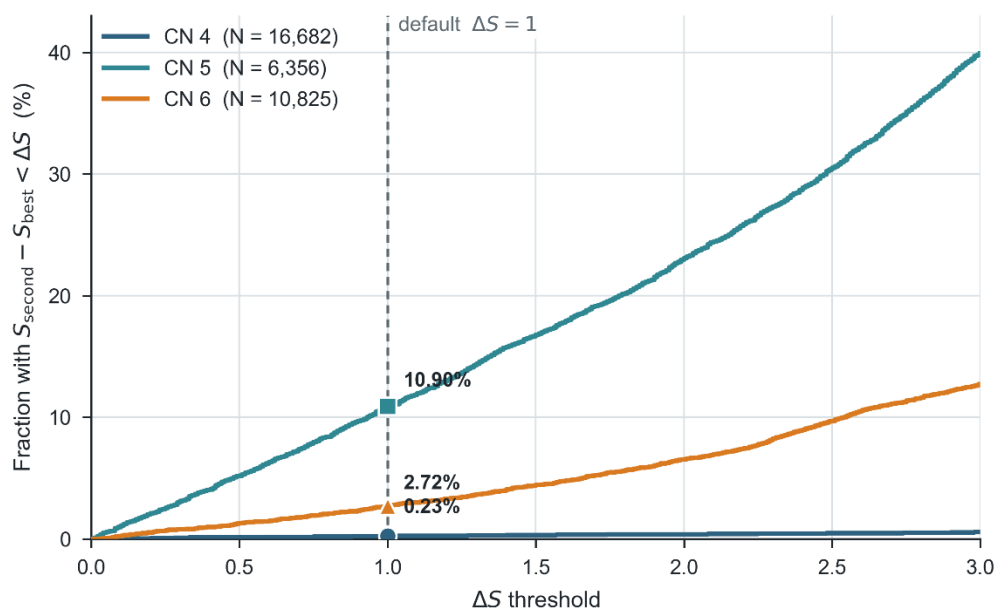
Condition	Per-donor Top-1 (%)	Exact donor set (%)
Full context	86.7	76.0
No geometry/stereo	86.5	75.7
Ligand masked	61.4	38.5
Ligand collapsed	58.4	47.7
Metal + CN only	60.3	37.4
Shuffled ligand	22.7	11.5

Supplementary Table S14B. Paired bidentate-conditioning test

Condition	Top-1 (%)
Partner visible	86.77
Both donors masked	86.67
Partner replaced	86.55

Partner-visible minus both-masked: +0.10 percentage points; 95% ligand-cluster bootstrap CI, −0.29 to +0.49; exact McNemar $p=0.804$. Exact donor-set recovery with both donors masked was 76.66%.

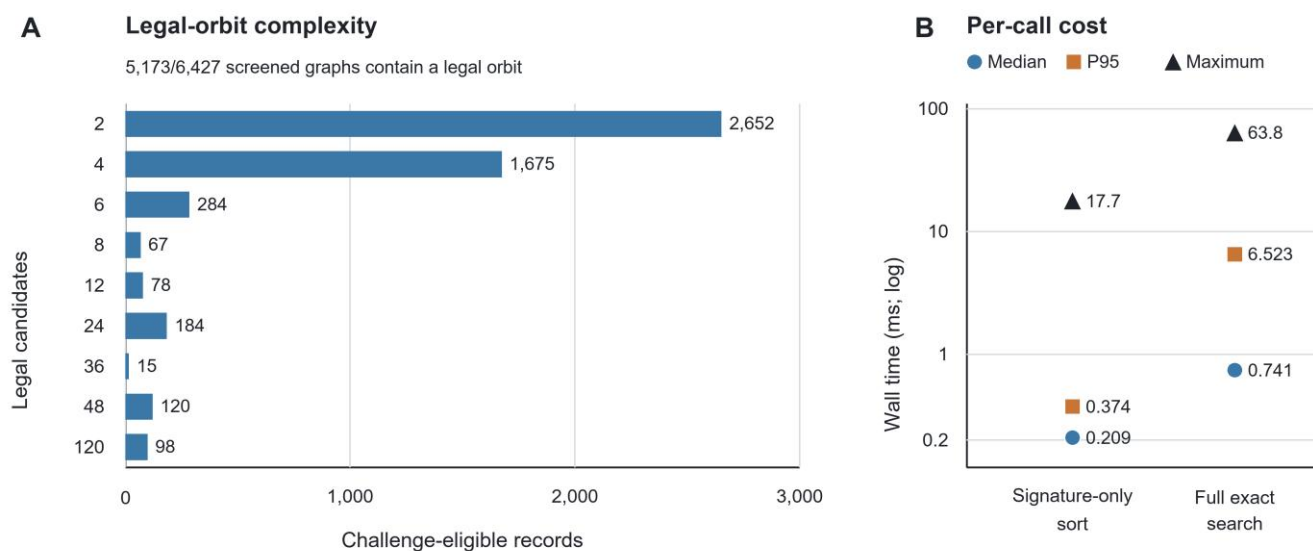
Supplementary Figures



Same corrected 33,863-record cohort as Figure 3; unrounded margins and a strict inequality were used.

Supplementary Figure S1. Sensitivity of the near-boundary fraction to the ΔS threshold.

Empirical cumulative fractions are shown for the same 33,863 corrected coordinate-space CShM records used in Figure 3 (CN = 4, $N = 16,682$; CN = 5, $N = 6,356$; CN = 6, $N = 10,825$). A record is classified as near-boundary when the unrounded margin $S_{\text{second}} - S_{\text{best}}$ is strictly less than the indicated ΔS threshold. Curves were evaluated from $\Delta S = 0$ to 3 in increments of 0.01. The dashed line marks the default $\Delta S = 1$, at which 39/16,682 CN = 4 records (0.23%), 693/6,356 CN = 5 records (10.90%), and 294/10,825 CN = 6 records (2.72%) are classified as near-boundary; the corresponding overall fraction is 1,026/33,863 (3.03%). The curves are empirical and unsmoothed. ΔS measures numerical proximity to the implemented ideal-shape references and is not interpreted as evidence for a reaction pathway or dynamic interconversion.



Supplementary Figure S2. Search complexity and computational cost of attachment-aware exact canonicalization. (A) Distribution of the exact legal-candidate count among the 5,173 challenge-eligible records identified from 6,427 unique graph-disjoint tmQMg/PBE CN = 4–6 snapshots. Candidate counts are the Cartesian products of symmetry-allowed equivalent-ligand and donor-rank-orbit permutations after chemical, attachment, and intrinsic-distance refinement. All eligible records remained below the frozen 100,000-candidate ceiling. (B) Descriptive wall time per canonicalization call for invariant-key sorting alone and the full exact minimum-whole-record search, reported as the median, 95th percentile, and maximum over 101,000 calls per method on the same machine. These timings are implementation diagnostics rather than a cross-machine performance benchmark. In the corresponding N = 1,000, K = 100 challenge, the full exact method produced 100,000/100,000 exact variant matches, 100,000/100,000 grammar round trips, stable candidate counts for 1,000/1,000 structures, zero variant errors, and zero false merges in the declared attachment control.

Source Data

All source data underlying the main-text figures, tables, and Supplementary Information results are provided in the frozen release/jacs-revision-20260823 directory of the CoordRep jacs-revision branch at <https://github.com/1701NJU/CoordRep/tree/jacs-revision/release/jacs-revision-20260823>. The accompanying manifests link each manuscript item to its CSV, JSON, or JSONL source, document the dataset and subset, and state the CSD redistribution status. Licensed row-level CSD structures, coordinates, and structure mappings are not redistributed.